Augmenting Microloan Risk Modeling through Data Synthesis and Transfer Learning: Interpreted with LIME and SHAP

By

Abrin Azad Era	ID: 22234103116
Trisha Majumder	ID: 21224103097
Murad Ahmed	ID: 21224103151
Md. Asif Al Azad	ID: 21224103171
Shema Akter	ID: 20215103051

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Declaration

We, hereby declare that the project report entitled Augmenting Microloan Risk Modeling through Data Synthesis and Transfer Learning: Interpreted with LIME and SHAP being submitted by us towards the fulfillment of the requirement for the course of "CSE 498B: Capstone Project", Department of Computer Science and Engineering is a project work carried by us under the supervision of Humayra Ahmed ma'am and submitted to her. We will be responsible if any mistake is found there.

Signature of Authors

Abrin Azad Era
ID: 22234103116

Trisha Majumder
ID: 21224103097

Md. Asif Al Azad
ID: 21224103171

Murad Ahmed
ID: 21224103151

Shema Akter
ID: 20215103051

Approval

I do hereby declare that the research works presented in this thesis entitled, Augmenting Microloan Risk Modeling through Data Synthesis and Transfer Learning: Interpreted with LIME and SHAP, are the outcome of the original works carried out by Abrin Azad Era, Trisha Majumder, Md. Asif Al Azad, Murad Ahmed, and Shema Akter under my supervision. I further declare that no part of this thesis has been submitted elsewhere for the requirements of any degree, award or diploma or any other purposes except for publications. I further certify that the dissertation meets the requirements and standard for the degree of Bachelor of Science in Computer Science and Engineering.

Supervisor

Humayra Ahmed

Assistant Professor

Department of Computer Science & Engineering

Bangladesh University of Business and Technology

Dhaka, Bangladesh

Chairman

Prof. Dr. Md. Ahsan Habib

Chairman

Department of Computer Science & Engineering

Bangladesh University of Business and Technology

Dhaka, Bangladesh

Dedication

Dedicated to our parents, teachers, friends and those who loved us for all their love and inspiration.

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Abstract

Although in emerging economies, information asymmetry is the major cause of Financial exclusion, the use of traditional credit scoring approaches is being limited by scarce data and reliance on informal sources of income. The paper gives a new three-step predictive computing model which overcomes these issues by shifting predictive intelligence used in data-abundant banking industries to data-poor microfinance industries. Our research model provides an elaborated system of integration that involves the integration of different microloan datasets with financial indicators to address the issues of data gaps. As the proposed Unified Risk and Eligibility calculation makes a clear distinction between fixed eligibility rules and the use of risk scoring using probabilities, risk and eligibility could be examined in more detail. Since no labelled default data are available, this paper makes use of transfer learning to adjust an Artificial Neural Network(ANN) that was initially trained on a US banking dataset of the Kiva microloan industry. This change is evaluated thoughtfully with the help of the Explainable AI (XAI) methods, that is, SHAP and LIME, which have to guarantee that the significance and visibility of the transferred financial pointers are preserved in the local economic setting. This strategy has a higher performance than a baseline model that is trained from scratch by having 89.52 percent validation accuracy and 86.95 percent overall accuracy. Also, the results indicate that the transfer of the developed banking knowledge to the microfinance-level setting is feasible and has a substantial positive effect on the accuracy of predictions and the level of transparency of the models.

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Chapter 1

INTRODUCTION

1.1 Introduction

Scholars believed inclusive finance to be crucial to sustainable development. However, it remains a difference between the developed and the developing countries. Advanced financial systems and credit reports that are stable in developed nations aid in loan evaluation. The process is also simplified by the modern digital technologies. A good credit history and well-structured paperwork is valued by borrowers. This streamlines the process of risk assessment. Therefore, developing countries have numerous challenges. These include ineffective financial records, unregistered income and inadequate technical infrastructure. These problems affected micro loans. They are also unavailable to formal financial services especially in rural economies and among women entrepreneurship. This minimizes their economic possibilities. Deep Learning and Artificial Neural Network have transformed decision-making in the financial industry. ANN models are able to work with massive financial data. They facilitate the identification of a connection between the loan default behaviour and credit risk. Transparency is enhanced by explainable Artificial Intelligence. It assists the borrowers and the lenders to interpret automated decisions. However, the majority of current ANN models rely on the information on developed nations, including U.S. bank loans. Informal employment, economic gaps, and poor credit history are some of the causes of failure of models in the developing regions. They perform well on the home front but not in foreign countries. The application of ANN-based deep learning in various fields of finance has serious limitations.

In this paper, a deep learning model is suggested. It applies transfer learning to limit such restrictions. A well performing ANN is required to be trained on bank data of the U.S. Thereafter, it modifies this model on the micro-credit datasets of the developing nations. A mixture of US bank loans, Kiva microloans, and local

socioeconomic profiles creates a definite dataset. A credible risk score is created by the financial, behavioural, and socioeconomic factors. This rating is used to show the personal risk levels of individual borrowers. The authors use XAI techniques, namely SHAP and LIME. These approaches facilitate openness, equality and equity in accessing finance. It is also better in forecasting loan approval and risks. Financial resources availability helps in developing borrowers especially in the low-income localities. Traditional banks are not likely to work in local realities where a large proportion of employment is informal.

In this paper, deep-learning is employed with XAI application to guarantee that lending is equitable, transparent, and consistent. The model is working in various regions and helps in development, business and reduction of poverty.

1.2 Purpose of the Study

The primary objective of this project is to create fair, clear and precise AI models in order to maximize loan appraisal in low-resource nations. The traditional credit models of the developing nations tend to be ineffective in these areas because of poor financial records, anomalies and local financial variations.

This study uses an effective model of the artificial neural network with explainable AI to assist financial banks in easy to comprehend and validate decisions. Using various data, including US bank loans, Kiva microloans, and regional loans data. The model is a depiction of client behaviour in a number of situations. It also provides a stable risk rating of individual clients, which makes the assessment of the client balanced and comprehensive.

The overall intention is to increase financial inclusion, offer equitable provision of credit, and offer information-based solutions that adapt to various financial circumstances.

1.3 Problem Statement

Loans are also important in promoting entrepreneurship and economic development as well as alleviating poverty. Nevertheless, the credit assessment process between

developed and developing countries is by far different. The developing countries have significant challenges of current loan evaluation procedures. The primary data sets involved in training the traditional AI models in loan risk and eligibility comes from developed nations, which in most cases are not a real picture of what is going on in undeveloped regions. Introducing AIbased modelling in these areas is heavily constrained by intermittent financial behaviour, incomplete paperwork, informal revenues, and economic instability in the areas. This leads to unfair and inaccurate judgments because future borrowers that may be credit worthy but do 14 not fit the credit profile are denied. In addition, most ML models are opaque, that is, make decisions without necessarily explaining them, and this causes borrowers to mistrust them and they cannot be adopted by financial institutions.

Consequently, the two essential issues that this study tries to solve are:

1. The question is how to construct machine learning models that can forecast appropriately when using multi-source datasets in developed and developing countries.

2. The method of making the credit evaluation process clear, clear and minimizes injustices and enhances financial inclusivity.

1.4 Research Objectives

The precise features of this work are to minimize the analytical and practical gaps in global loan evaluation systems. Among the goals are:

- Stable machine learning models that predict credit risks and eligibility of US. bank loan datasets with a high precision

- Detecting the microloan issuing through Kiva data collections and area based loan themes and country-specific descriptions, which reflect a variety of financial trends and behaviours.

- Determine useful attributes in the datasets to form a unified structure of analysis capable of learning how borrowers behave under various financial situations.

- Using methods of XAI, such as SHAP and LIME, to guarantee the intelligibility of results of the system.
- Promotion of financial integration through the creation of the models that have taken into account the different socio-economic characteristics, situations in the regional economies, and informal income sources

Overall, these goals will have the study present explainable, fair, and context sensitive means of credit evaluation in developing regions as well as valid models.

1.5 Research Contribution

The study contributes greatly to the areas of financial inclusion, machine learning, and financial assessment:

- **Increased Predictive Accuracy:** The structure gives improved prediction accuracy on heterogeneous datasets by combining microloan data and transfer learning based on U.S. bank datasets.
- **Lessening Data Gaps:** A unified logical structure is developed through integrating information on developed and developing nations, allowing the knowledge to be transferred into both formal and informal financial practices.
- **Open Decision-Making:** The use of XAI strategies promotes accountability by making sure that the financial givers and borrowers can comprehend why certain loans were chosen.
- **Financial Modelling with Inclusivity:** The paper encourages equal access to loans to disadvantaged populations by addressing unofficial revenue, local variations, as well as non-standard financial practices.
- **Practical Use:** Banks, microfinance organizations, and fintech platforms can practically implement the frameworks since they have been designed to be computationally efficient as well as appropriate for actual financial platforms in real-time.

The research offers holistic and socially acceptable methods of minimizing global financial imbalances due to such contributions.

1.6 Motivation

This finding is caused by the consequences of loans, particularly, microloans. Personal loans approval rates usually stand at between 20 to 40 per cent as compared to SME approval rates which range between 30 and 50 per cent. An organised credit system would go a long way in enhancing the approval rate of loans to both Personal and SME loans like in the developed countries. Nevertheless, the scarcity in economics and microloans would result in the highest 16 approval and utilisation rates because of highlighting their value in the development of financial services and entrepreneurship. The problem is exacerbated by the inaccessibility of sophisticated analytical materials in the poor neighbourhoods. Those borrowers who have succeeded have changed their credit system which is traditional credit, to formal credit system. Conventional models of loan evaluation cannot consider non-standard patterns of income distribution, culture and economic instability of the region. In this study, the authors demonstrate the necessity of just, verifiable, and data-driven approaches considering the socioeconomic variables and access to equal credit.

1.7 Flow of the Research

The research has a framework of a workflow that helps to ensure that there is coherence and clarity all throughout the investigation. The first steps in the process are identification of the issue, purpose and goals of developing a cross-regional loan assessment system. The second step is the collection, pre-processing, and merging of datasets of regional loan themes, nation profiles, Kiva microloans, and U.S. bank loans. To model the borrower-specific risk profiles, model development is subsequently done using the recently developed risk score, using the machine learning and transfer learning algorithms.

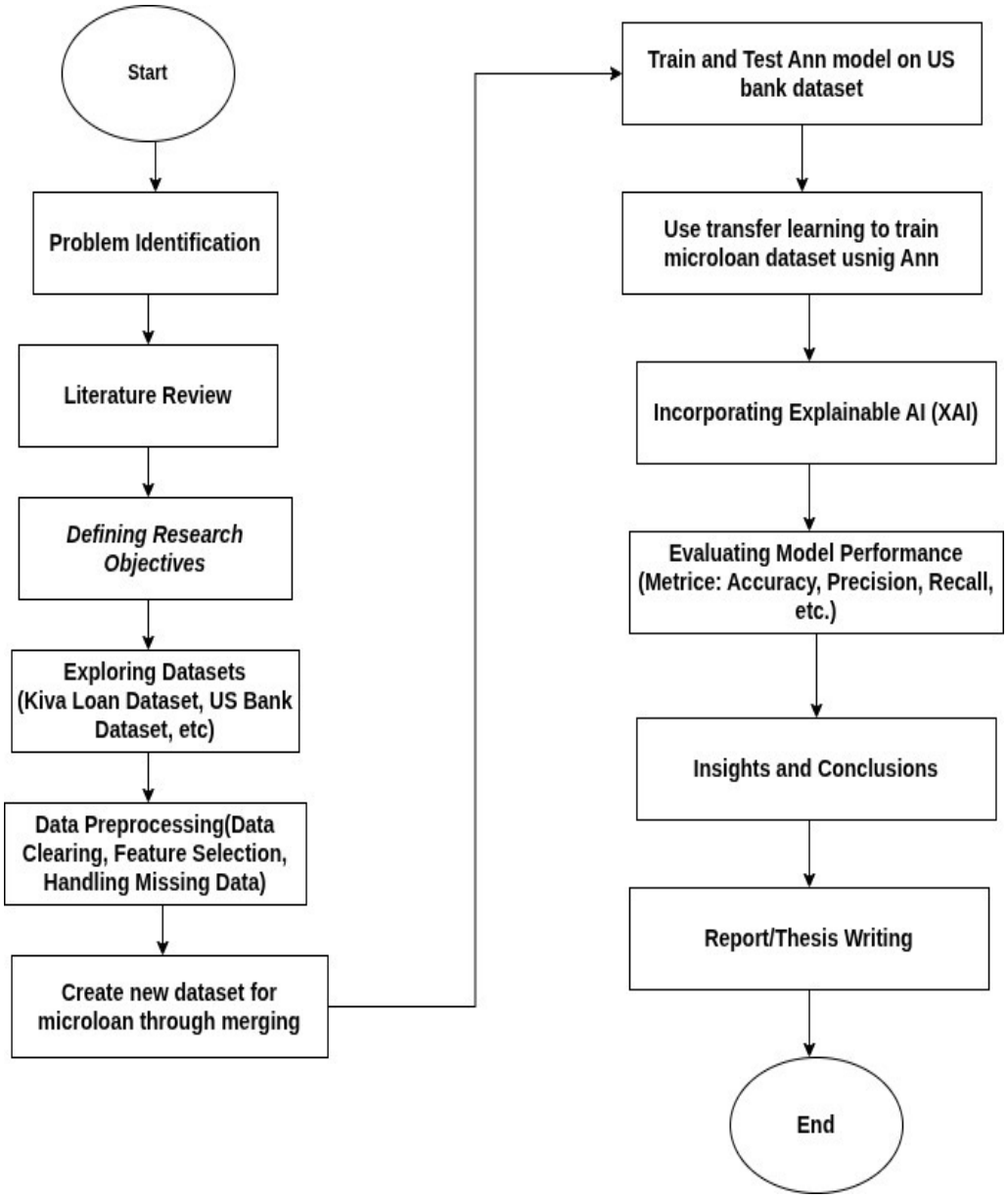

Figure 1.1: Flow of the Research

The models are then assessed by standard measures such as accuracy, precision, and recall, F1-score, and con- confusion matrices. Finally, explainable AI methods are also used to interpret model predictions, which ensures fairness and transparency. The conclusion of the workflow offering performance evaluation, discussion, and further improvement recommendations can be considered as a comprehensive structure of a piece of scholarly research and practical implementation.

1.8 Summary

In this chapter, the background of the study, its purpose and role are introduced. It deals with the issues of loan eligibility and credit risk analysis in the developing countries. The significance of machine learning and explainable AI in diminishing the disparity in the developed and emerging financial systems is also highlighted in the chapter. Thus, it presents the research problem, objectives, motivation, contributions, and research process, which in combination offer a clear background to the chapters to come. In the following parts of this 18 report, the essence of the cap- stone project is described such as the description of the dataset, the research methodology, the model development, the evaluation, and the analysis of the results. With the control of both predictive and interpretating accuracy, this project will create an efficient, inclusive, and socially conscious loan evaluation frameworks that will promote financial inclusivity and sustainable economic progress within.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

Credit risk assessment and loan eligibility review are used by banks to control the default risk and revenues. Conventional approaches including logistic regression are good but, in most cases, do not capture non-linear and complex trends in client financial data. AI models enhance the model accuracy through large and diversified datasets but most of the models are black boxes, thus, diminishing confidence in credit decisions. Strategies like SHAP and LIME are explainable AI and can be used to overcome this problem by making predictions clear. Interpretability is highly important in small loans cases where the decisions are made with little or irregular data on communities that are not in the financial centres. This chapter provides the manner in which deep learning networks and explainable AI could be merged to enhance objective decision making, insight, and efficiency in credit risk identification and loan issuance assessment.

2.2 Literature Review

To forecast credit risk, the proposed Logistic Regression model based on the Random Forest can be integrated with explainable AI and the high-performance classifier together. It has 93% predictive accuracy and enhances transparency with feature-level explanations. Notwithstanding, it suffers problems with scalability to larger datasets and data generalizability. To ensure fairness and legal adherence, the authors justify explainable AI in the financial sphere [1]. In addition, the Decision Trees algorithm classifies the XAI approaches applied in finance by aiming to increase the trust in the AI decisions and strike a balance between interpretability and accuracy. In the case of financial decision-making systems, their structure is suggested [2]. Even though it can only be applied to English reports, the FinBERTXRC method, trained on 10-K

reports using XGBoost, is more predictive and through increasing explanations. It is recommended as risk management [3]. Although the accuracy concerns are admitted, the studies based on Lending Club data highlight the benefits of ensemble models and neural networks as credit grading tools. To ensure trust in financial services, explainable machine learning is suggested [4]. Another analysed Lending Club data is also a modern version of a Random Forest which captures the advantages of ensemble models, but does not take integration issues into account when considering the large banking systems [5]. Finally, despite having the problem of scalability, an XGBoost model based on the Norwegian bank data reveals the need to have a certain level of explainability to enhance transparency and compliance [6]. To predict creditworthiness, Explainable AI incorporates explainability techniques like SHAP and LIME with machine learning models in credit risk analysis. Algorithms such as Random Forest and Logistic Regression enhance transparency but with concerns regarding evaluations and recommendations of finance services and provide fairness, still rely on factors such as computational costs and interaction of multiple features [7]. Random Forest is the most accurate among the models examined and used in loan acceptance such as the Decision Trees and the Support Vector Machines. Due to its simplicity, the Logistic Regression often performs better than other models, which explains the importance of model interpretability to regulatory compliance [8]. To determine the likely defaulters, the Random Forest Regressor will be utilized to forecast the loan approvals with regard to such variables as gender, employment, and education. It is stated in the study that other customer attributes, other than income can be used to enhance prediction accuracy [9]. A comparison-based on a Kaggle dataset comprising of 10,128 instances revealed that logistic regression was the most accurate at 92%; however, data imbalance and dynamism of financial data is mentioned [10]. Although the authors acknowledge the risks of overfitting and imbalance of data, they support the idea of incorporating machine learning in faster and more objective decisions on loans [11, 12]. The paper also considers counterfactual explanations in credit scoring systems, where minor manipulations in the characteristics of the applications can give varying results. The

authors develop a cost function to develop believable and realistic counterfactuals. This all-encompassing strategy aims at enhancing the effectiveness of generating such explicative [13]. Lastly, the use of AI in credit risk evaluation is discussed, and it demonstrates how various AI models could be applied to analyse big data to enhance the accuracy of default prediction. AI enhances efficiency in the banking sector and facilitates ethical lending behaviour by automating banking and tailoring loan offers. However, the research also brings up problems, such as the need to use AI with ethical intentions in the financial field and the influence of biased data on outcomes. To enhance the credit risk assessments and optimize the banking processes, the authors recommend the continuous improvement of AI techniques [14]. With XAI methods such as SHAP and LIME, a financial institution can enhance the explainability of their models, making them more regulatory-compliant and transparent without loss of predictive accuracy. Despite the fact that black-box models such as deep learning and ensemble techniques are much more effective in such tasks as fraud detection and credit scoring, their inability to be interpreted raises ethical and regulatory concerns [15]. To ensure that the decisions made can be explained and comply with the regulations of such laws as GDPR and address the issue of algorithmic bias, the study encourages the implementation of XAI. Moreover, it recognises constraints like scalability and complexity of understanding decisions alongside focusing on the application of explainable methods to make clear predictions in credit scoring especially using unbalanced datasets [16]. Research shows that based on racial biases in computerized mortgage underwriting, Black applicants should expect more loan denial and higher interest rates than their similar white applicants. Such a bias is caused by the fact that historical evidence indicates racial imbalances in the mortgage industry. A study based on the 2022 HMDA dataset found that black applicants were required to score nearly 120 points higher than white applicants on their credit score, in order to get accepted. It is interesting to observe that this bias can be reduced by requesting models to disregard race in decision making. It is underlined by the fact that AI-driven operations should be fair since such laws as the Fair Housing Act and the ECOA prohibit any discrimi-

nation when lending money [17]. Explainable AI (XAI) is also being applied to the financial services and banking sector to enhance transparency of decision-making, particularly in regard to loan approval. Traditional machine learning models often work as black-boxes, forcing them into consideration of their compliance with other rules of fairness such as the ECOA. XAI guarantees compliance with regulations by applying algorithms such as SHAP and LIME to make the processes fairer and more transparent, thereby improving a deeper insight into the logic behind the approvals or rejections of loans [18]. Transparency is essential to gaining the trust of customers, according to a study on AI in credit risk assessment. It talks about interpretability methods that improve comprehension of AI-based choices utilizing public credit datasets, such as LIME and SHAP. The study emphasizes the difficulties in implementing these techniques across various datasets and promotes explainable AI frameworks to increase confidence in credit evaluations [19]. In order to enhance credit risk predictions, the paper "Effective Credit Risk Prediction Using Ensemble Classifiers with Model Explanation" advocates for ensemble classifiers such as Random Forest, XGBoost, AdaBoost, and LightGBM. Although their complexity makes training and interpretation difficult, these algorithms improve accuracy and interpretability by integrating numerous models, especially when employing German and Australian datasets [20]. To improve loan approval prediction accuracy, a separate study investigates the classification of loan applications using bio-inspired neural networks [21]. According to a different study, LightGBM may reach a 96.2% accuracy in credit risk assessment for rural microfinance by correcting data asymmetry and assessing variables including income and payback history [22]. Last but not least, Explainable AI (XAI) approaches are becoming more and more popular in the banking industry, with an emphasis on model-agnostic methods for customer segmentation and credit risk assessment. Well-known techniques like SHAP and LIME are mentioned, however there is a lack of clear explanations for those who are not experts. By using these techniques, AI-driven financial decisions may become more transparent [23, 24]. The research provides one method of reading credit default predictions, which involves a Light Gradient Boosting Machine algorithm

and SHAP that enhance the credit scores transparency by identifying the significant ones. It depends on one data set thereby restricting its applicability despite its better results compared to the traditional models. The authors stressed that financial organizations can comply with explainability conditions in AI-based financial decisions by integrating XAI methods [25, 26]. A further ensemble machine learning model is also introduced to improve prediction of credit risk with a high degree of interpretability. It recognizes significant factors and visualizes decision making process using a combination of diverse base models and interpretation methodologies. It is reported to make lending applications more trustworthy and the approach performs better than base models in accuracy [27]. Another paper is on the area of data mining techniques and pre-processing of data to address the issue of missing data, using the K-Nearest Neighbour (KNN) algorithm to classify credit loans eligibility in cooperatives. Performance suffering with unimportant features is possible with KNN, even when used on noisy data. The authors suggest that their enhanced framework would enhance decision making in credit evaluation [28]. To enhance the loan approval prediction based on such algorithms as Random Forest and Logistic Regression, ensemble learning and feature selection are also examined in the study. Though it emphasizes the challenges of uneven datasets that lead to biased forecasts, it also emphasizes higher accuracy using the methods of ensembles. Banking systems are recommended to implement this in order to enhance the processes of loan approval [29].

2.3 Summary

In recent years, research indicates that machine learning techniques, particularly those based on ensemble models, can provide better results when they are applied to loan approval and credit risk classification problems. Many of these works are based on practical issues, including the existence of the imbalanced data, varying data quality, and the problem of selecting the most useful features to obtain superior predictions. Another point that is frequently emphasized by the researchers is that interpretable models are significant due to the ease with which they make financial

decisions easier to justify and comprehend. Overall, this body of literature suggests that machine learning-supported and data-driven methods are increasingly popular in the current credit risk assessment.

Chapter 3

PROPOSED MODEL

3.1 Introduction

Eligibility of loans and credit risk evaluation is a significant undertaking in the financial services sector. Nevertheless, the classical approaches tend to ignore the important variables in less developed or emerging markets. This chapter presents a model that integrates Explainable AI (XAI) and ANN to bridge the developed and developing credit environments. This research uses two different datasets: a US Bank Loan dataset, where there is ample data to learn simple risk patterns is supplied, and a Kiva Microloan dataset, where there is little information to assess risk in traditional ways is supplied. This research applied the method of Transfer Learning and Domain Adaptation to the proposed model that will guarantee the introduction of fairness and clarity to decision making. Moreover, give useful information regarding the qualification and risk forecasting in various economic environments.

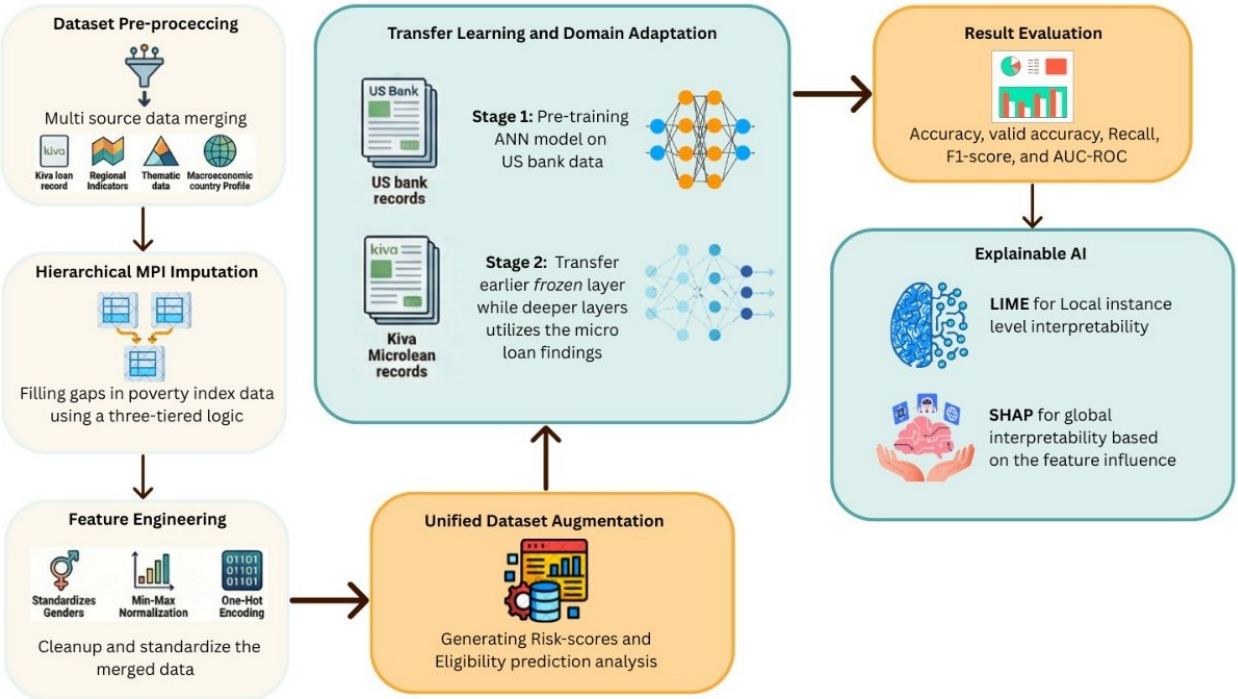

Figure 3.2: Outline of the proposed model

3.2 Dataset

Our data were obtained through two different financial sources, one of them is a traditional US Bank, which is provided by Kaggle, and the second one is Kiva, which is a microfinance platform worldwide. The data of the US Bank represents a clear and high-resource baseline, whereas the Kiva data reflect the informal and complex reality of the micro-lending. The two datasets provide useful data on loan eligibility and risks.

No	Dataset Name	Columns	Rows
1	US Bank Loan Dataset	37	20,000
2	Kiva Country Profile	38	86
3	Kiva Loan Theme by Region	16	15,763
4	Kiva MPI Region Location	9	1,009
5	Kiva Loans	19	6,71,205

Table 1: Datasets Used in The Research

The data utilized in this research can be presented in Table 3.1, where the comparison of the size and structure is also presented. The US Bank Loan data is a normal, organized data normally found in the lending system of a developed country. Conversely, Kiva Microloan data is spread over multiple unique files to reflect the various levels of microfinance illustration country level economics, poverty themes of the region, geo-spatial data at specific location and individual loan details. Combining these subsets together we take personal borrower information and combine it with the bigger regional and national picture.

I. Dataset 1: (US Bank Loan Dataset)

20,000 loan application records gathered from a reputable financial institution make up the US Bank Loan dataset. It includes a variety of financial, demographic, behavioural, and credit related factors that describe the income, assets, obligations, employment stability, credit history, and loan characteristics of applicants.

Applicant Demographics			
Age	Applicant's age	MaritalStatus	Marital status
EducationLevel	Highest education level	NumberOfDependents	Number of dependents
EmploymentStatus	Employment status	JobTenure	Years at current job
Experience	Total work experience (years)	HomeOwnershipStatus	Home ownership status
Income and Financial Profile			
AnnualIncome	Annual income	MonthlyIncome	Monthly income
SavingsAccountBalance	Savings account balance	CheckingAccountBalance	Checking account balance
TotalAssets	Total assets	NetWorth	Net worth
TotalLiabilities	Total liabilities	MonthlyDebtPayments	Monthly debt payments
UtilityBillsPaymentHistory	Utility bill payment history	TotalDebtToIncomeRatio	Debt-to-income ratio
Credit History and Behavior			
CreditScore	Credit score	LengthOfCreditHistory	Length of credit history
NumberOfOpenCreditLines	Open credit lines	NumberOfCreditInquiries	Recent credit inquiries
CreditCardUtilizationRate	Credit card utilization rate	PaymentHistory	Past payment behavior
PreviousLoanDefaults	Previous loan defaults	BankruptcyHistory	Bankruptcy history
Loan Details			
ApplicationDate	Loan application date	LoanPurpose	Purpose of loan
LoanAmount	Requested loan amount	LoanDuration	Loan duration (months)
BaseInterestRate	Base interest rate	InterestRate	Applied interest rate
MonthlyLoanPayment	Monthly repayment amount	DebtToIncomeRatio	Debt-to-income ratio
Outcome Variables			
LoanApproved	Loan approval status	RiskScore	Computed risk score

Table 2: Attributes of US Bank Loan Dataset

These characteristics facilitate thorough evaluations of loan eligibility and credit risk. In this work, the dataset serves as the source domain for deep learning models that are then trained using transfer learning and domain adaptation approaches to adapt to microloan data.

II. Dataset 2: (Kiva Microloan Dataset)

The second set of data changes to the developing world. This dataset is a differentiated group of several subsets. A data engineering pipeline is used to unify the Kiva data.

i. Kiva Loans (The Core Data): It is the biggest subset, which comprises 671,205 records. It is the primary ledger which captures the level of transaction like

Loan Amount, Repayment Interval, Sector, and Lender Count. This data will be the one that tells us who is borrowing and the reason why they are borrowing.

ii. Kiva Loan Theme by Region: This theme contains 15,737 records in 21 columns and relates loans with their geographic and thematic settings. Therefore, it assists us in better comprehending the purpose of the loan to be used in the particular frameworks of specific regions.

iii. Kiva MPI Region Locations: It is a geo-spatial dataset which comprises 1,009 records and 9 columns to precisely map the Multi-Dimensional Poverty Index (MPI) to particular locations. The mapping assists in a comprehensive evaluation of the economic vulnerability at the regional level.

iv. Kiva Country Profile: It is a significant indicator that gives a broader perspective of the economy of the country consisting of 86 records and 38 columns. It associates the economic indicators of national significance such as GDP Growth and Employment Structure with the country of the borrower. This enables the model to change risk scores according to systemic and economic factors.

Gathering these sources and matching them with common keys allowed transforming sparse individual records into a context-filled new dataset. This combination matters to what is referred to as the Transfer Learning process. It enables the model to apply the banking logic within the context of microfinance.

3.3 Methodology

In this project, the implementation of the artificial neural network and Transfer Learning is done using Deep Learning libraries, whereas pre-processing and feature encoding is done with the help of standard data science tools. We incorporated explainable AI methods, including LIME and SHAP to render the model explainable. Because the data required to perform the risk and eligibility analysis was widely spread on the various datasets, consolidation was an important process in my project. Through the synthesis, I generated one dataset that encompassed the borrower data, sector data, country economic indicators, regional poverty rate (MPI), and the theme of loans in each record of the loans. Risk scoring and machine learning

prediction would not have been possible without merging as the data is not enough. Hence, merging also guaranteed that all the useful features were included in a single set of data that can be analysed, and that became the basis of the project.

I. The Merge Process and Dataset Integration

The initial stage of this study was Data integration. To obtain the desired variables to be used in a full-fledged risk assessment, which were distributed across various Kiva datasets, we applied various merging techniques. We joined records based on such keys as Region and Country, which provided us with a single dataset that provides the context of each loan, the details of borrowers, and risks and macroeconomic indicators of the sector. This step would have helped the model to overcome data sparsity. The merge will make sure that the borrowers possess valuable contextual features.

II. MPI Cleaning and Integration

One of the major indicators of vulnerability of the borrowers and environmental risk is the Multi-Dimensional Poverty Index (MPI). Nonetheless, the raw Kiva data were inconsistent in the naming of regions and MPI values gaps. To correct this, we conducted an extensive preprocessing that emphasized on the loan themes by region and the MPI region locations datasets.

Pre-processing and Cleaning: We have eliminated any unnecessary columns in both data frames. This served to decrease dimensionality as well as concentrated on pertinent geo-spatial identifiers.

Primary Merge: We had merged datasets by the mpi-region and LocationName, but we kept the MPI column, so that we did not have duplication of features.

Hierarchical Imputation: Our logic applied to fill in the null MPI values in order to enhance data completeness was as follows:

- Textual Match: In case MPI was considered to be null, the algorithm tested whether the names column contained any words that were a match of a region. In case it matched, it occupied the cell that was left empty.

- Region Match: In case the MPI had been null, the system verified whether the region column was an exact match of the region column.
- Location Match: Lastly, in case the MPI was not the case, the system verified whether the LocationName was equal to the LocationName. When they were a match, it occupied the vacant cell.

This stratified method also implied that the maximum number of borrowers received the right regional poverty score to remove the possibility of the model of getting rid of the potential viable applicants due to the unavailability of some data

III. Pre-processing and Feature Engineering

Both the US Bank and Kiva sources of the raw data had to undergo significant transformation to get machine learning ready.

i. **Feature Engineering:** Since Kiva raw data were not consistent, fragmented, and not suitable to use in machine learning, feature engineering played a critical role in my strategy. In order to convert the raw data into useful features and create an effective risk score and a prediction model of eligibility, we engineered the features by standardizing genders of the borrowers, scaling the numerical data, coding the categorical variables and coming up with other indicators such as the vulnerability indicators, the Loan to GDP ratio and the Loan to MPI ratio. These manufactured features allowed the model to understand better the borrower behaviour, financial capability, and the socioeconomic risk. Without proper feature engineering, the ANN model, risk scoring method and eligibility forecasts would be inaccurate and unreliable.

ii. **Normalization & Encoding:** The one-hot encoding was applied to the categorical variables (e.g., Gender, Sector). Min-Max normalization was used to scale the numerical values with the aim of rendering the Neural Network to converge effectively.

iii. **Outlier Management:** Outliers and noise have been handled to avoid the skew of the model especially in the mixed microloan data.

IV. Unified Risk and Eligibility Architecture

The data obtained is then processed to our Unified Architecture. This plan does not fit the typical dichotomy of credit rating. The risk assessment and the eligibility criteria are normally lumped together in the systems of older systems. It can result in ambiguous rejection. Our framework splits these into two separate layers:

i. **Eligibility Engine (Deterministic Layer):** This layer is a hard filter on institutional policy, donor requirements and regulatory provisions. It verifies Boolean criteria, including region validity, sector compliance and age limit, before some complicated calculations take place. This guarantees that applicants who go badly wrong are filtered out.

ii. **The Risk Scoring Engine (Probabilistic Layer):** The layer is based on the probabilistic output of our ML model to generate a finer risk score (0-100), or Probability of Default (PD).

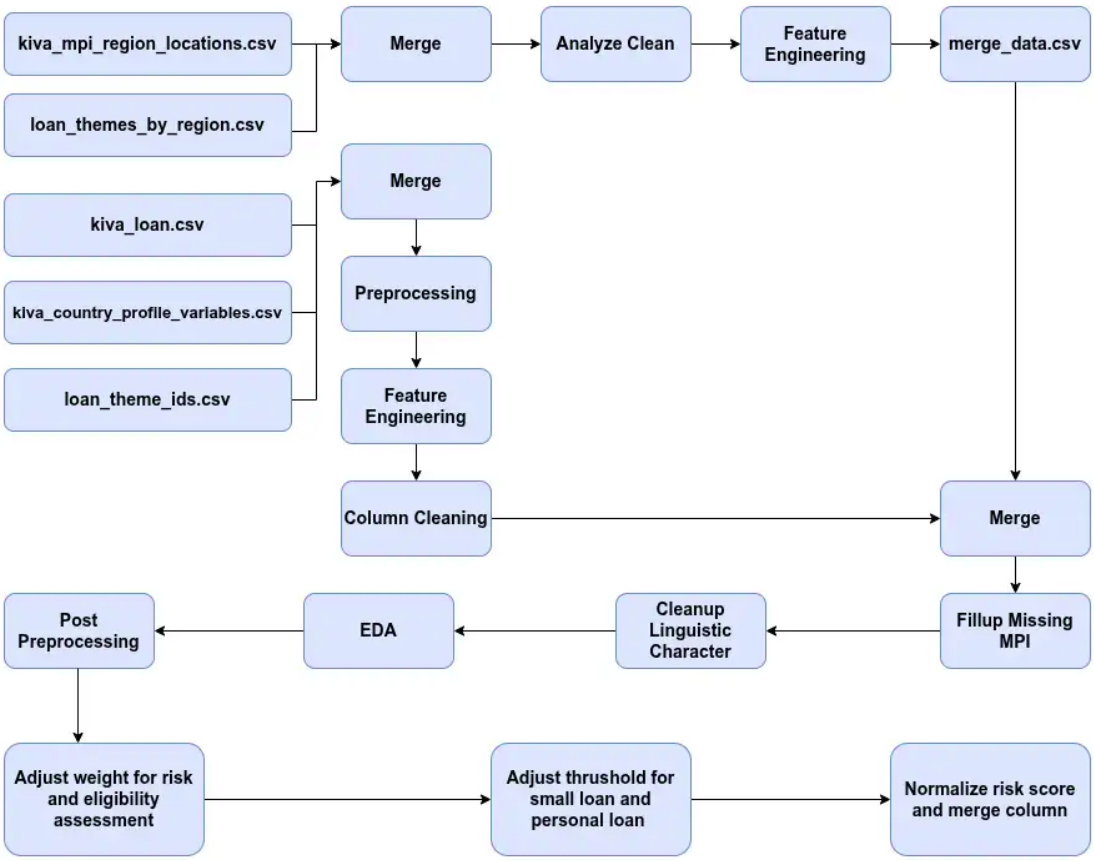

Figure 3.3: Data Augmentation for a unified Kiva dataset

The system allows the Microfinance Institutions (MFIs) to tailor their risk appetite independently of modifying their fundamental eligibility policies and offer a flexible program to manage portfolio exposure.

V. The Artificial Neural Network (ANN)

ANNs have performed well in the modelling of nonlinear relationships that are complex in financial data. We adopted ANN as a primary prediction model and another one as a trainer model to conduct the evaluation of credit behaviour and loan eligibility. Our first approach involved the training of the network on a standard bank loan dataset to determine some basic customer credit history trends and repayment behaviour. The ANN may be able to identify latent risk patterns that may not be identified by other models due to its layered learning model. This may result in better predictions that will be used to approve loans and their likelihood of being repaid. Consequently, the inclusion of the ANN in the microloan riskassessment system enhanced the effectiveness and reliability of the predictions to a great extent.
Mathematical Formulation: The computational activities of an ANN as well as learning can be mathematically modelled as follows.

i. **The Weighted Sum (Linear Combination):** This is the first step where the model takes input features and assigns them importance using weights.

$$z = \sum_{i=1}^{n} w_i x_i + b \tag{1}$$

   - x_i : input data (features).
   - w_i : The weights (the "importance" the model gives to each feature).
   - b : The bias (allows the model to shift the activation function left or right).

ii. **The Activation Function (Sigmoid):** The value z can be any number from negative to positive infinity. To turn this into a probability (between 0 and 1), we pass it through the Sigmoid function.

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} \tag{2}$$

- If z is a large positive number, $\hat{y}$ approaches 1.
- If z is a large negative number, $\hat{y}$ approaches 0.

iii. **Binary Cross-Entropy Loss (for training):** During training, we need to measure how well the model is performing. This formula calculates the "cost" or "loss."

$$Loss = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{3}$$

- y_i : The actual label (0 or 1).
- $\hat{y}_i$: The predicted probability from the Sigmoid function.
- How this works: If the actual label is 1 and prediction is close to 0, the loss becomes very high, signalling the model to adjust its weights.

VI. Transfer Learning

Traditional machine learning methods assume that training and test data come from similar feature spaces and distributions. However, this assumption does not hold in the context of global financial inclusion. The lack of labelled default data in developing markets creates challenges. Using transfer learning takes knowledge gained from solving a problem in a resource-rich source domain and applies it to improve performance in a resource-constrained target domain. Specifically, the source domain is US Bank Data with the learning task of General Credit Risk Scoring. The target domain is Kiva Microloans with the learning task of Micro-credit Eligibility. Transfer learning aims to enhance the predictive function for the target by using the knowledge obtained from the source.

Methodological Approach: This research uses Inductive Transfer Learning. The target task, predicting micro- loan defaults, is different from the source task. However, we have labelled data in the target domain that helps refine the model. We specifically apply a Parameter-Based Transfer method. Training Artificial Neural Network (ANN) models from scratch often results in overfitting, especially with small datasets. To avoid this, we reuse the pre-trained weights from the US Bank

model, which also helped us to validate the columns used, even if the context was different. The learning process istwo-phase process:

- **Pre-training (Source Domain):** The initial ANN model is trained on the initial US Bank data to get to know general features of credit risk and the relationships between columns. These involve the knowledge of the non-linear interaction between leverage ratios and the repayment probability.
- **Fine-Tuning (Target Domain):** Here the layers that form the first stages of the network containing major financial behaviour are maintained frozen. We then re-initialize and train the deeper network with Kiva dataset. It enables the model to have core banking expertise as it modulates its decision scope to socio- economic aspects

VII. Domain Adaptation

Whereas Transfer Learning assists in data scarcity, Domain Adaptation addresses the disparities in feature distributions of the developed and developing world conditions, such as FICO scores versus social reputation. Once the transfer stage is completed we apply feature- based adaptation methods to map the various features into a shared latent space. It ensures that the structural knowledge of the US Bank setting remains valid mathematically and economically in the diverse Kiva microloan setting. Heterogeneous Transfer Learning is applied to address the challenge of the Cold Start problem in microfinance, whereby little or no historical realization data is available. In this approach, the ANN is trained with the US Bank Loan data (Source Domain) which contains lots 35 of data to learn the universal latent features of credit risk such as the non-linear correlation between debt burden and repayment probability. After the model is trained on the source data, we freeze the weights of the early stages of the model to retain this critical financial knowledge. The model is then optimized with the Kiva Microloan data (Target Domain) which in effect enables the model to adapt to the particulars of micro-lending without commencing at the beginning

3.4 Evaluation and Explainability

Our unified micro-loan data set was over-fitted and lost data and to mitigate this and other issues, we conducted a two-stage analysis. The explainable AI was used to create transparency on the logical line

I. Evaluation Metrics

The performance of the model was evaluated by four standard measures, namely accuracy, precision, recall, and the F1-score measure. Whereas recall gives the degree to which the model was able to identify the presence of real positive scenarios, precision measures the amount of positive cases which were predicted to have been accurate. The total percentage of correct predictions is known as its accuracy, F1-score provides a balanced score, which is a combination of Precision and Recall. Moreover, the confusion matrix was used to summarize true positives, true negatives, false positives, and false negatives, which was used to identify a clear picture of the performance by classes.

i. **Accuracy:** The easiest measure, Accuracy is used to determine the percentage of correct guesses made by a person across all the attempts.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

ii. **Precision: Precision answers:** What portion of the time has the model been correct in saying that something was Positive? It is about reliability.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

iii. **Recall (Sensitivity):** Recall answers: "Of all the actual Positive cases that exist, how many did the model catch?" It is about completeness.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

iv. F1-Score: Because there is usually a "tug-of-war" between Precision and Recall (improving one often hurts the other), the F1-Score provides a single balance point using the harmonic mean.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

Expanded Form:

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{8}$$

II. Explainable AI (XAI)

Explainable AI (XAI) will be significant in order to make the loan and microloan risk and eligibility models more comprehensible, understandable, and reliable. XAI incorporates the techniques that demonstrate how predictions are affected by inputs and weights and the complex machine learning models are simplified. XAI techniques were employed similar to SHAP and LIME to describe the choices of the Artificial Neural Net- work (ANN) models. This enables the stakeholders to interpret, confirm and believe in the results of the model since they can know which attributes, among others, in terms of loan value, MPI values, country of borrower, and previous track record of loan repayment contributed the most in the predictions. This transparency is particularly important in the field of finance where the decisions are required to be transparent, equitable, and reasoned out. Consequently, XAI enhanced transparency, assisted in detecting possible biases, and generated valuable information in the case of both conventional loans and microloans.

III. SHAP (Shapley Additive explanations)

SHAP is applied to offer Global Interpretability. The SHAP values are based on the cooperative game theory and indicate the input of each feature to the ultimate prediction of the model. This allows us to evaluate the Transfer Learning process itself: when the transferred model relies heavily on the features that are not essential in the microfinance setting, SHAP analysis will reveal this discrepancy, which will signal the Domain Adaptation failure. It makes sure that the model makes right

decisions. SHAP was highly beneficial to this project and it was evident that the features had a positive impact on the forecast of loan and microloans risks. SHAP contributed to determining the most influential criteria in making eligibility decisions by allocating the values of the importance of such features as the loan amount, MPI, borrower nation, and payback history. This promoted more trust and accountability as the predictions of the model were comprehensible and reasonable and in line with financial judgment.

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{9}$$

- $f(x)$: The actual prediction made by the model for a specific input.
- ϕ_0 (Base Value): The "starting point." This is the average prediction across the entire training dataset (what the model would predict if it knew nothing about the specific features).
- ϕ_i (Shapley Value): The specific contribution (positive or negative) of feature i to the final prediction.
- M: The total number of features.

IV. LIME (Local Interpretable Model-agnostic Explanations)

Besides SHAP, there is also Local Interpretable Model-agnostic Explanations (LIME) which relies on interpretable and locally approximated models to interpret particular predictions. The case-level judgments were further clarified by LIME to analysts by underlining the salient characteristics of each borrower that determined a particular level of eligibility or risk score. SHAP and LIME together are more interpretable, both locally and globally, improving transparency, determining possible biases, and enhancing the strength and reliability of the risk and eligibility models even when working with a diverse set of data on microloans. It is similar to SHAP in that it gives Local Interpretability. SHAP justifies the model, whereas LIME justifies the instance. It is an approximation of the non-linear, complex ANN with a simple and interpretable linear model around a given loan application. This allows one to gener-

ate human readable explanations of individual borrowers. Furthermore, granularity is critical to the provision of equity and down-to-the-ground feedback to unbanked applicants.

3.5 Proposed Model

The proposed model realizes the study results in three steps, well-integrated architecture, which moves the data fusion step to a unified evaluation step, and then to an Explainable Transfer Learning engine.

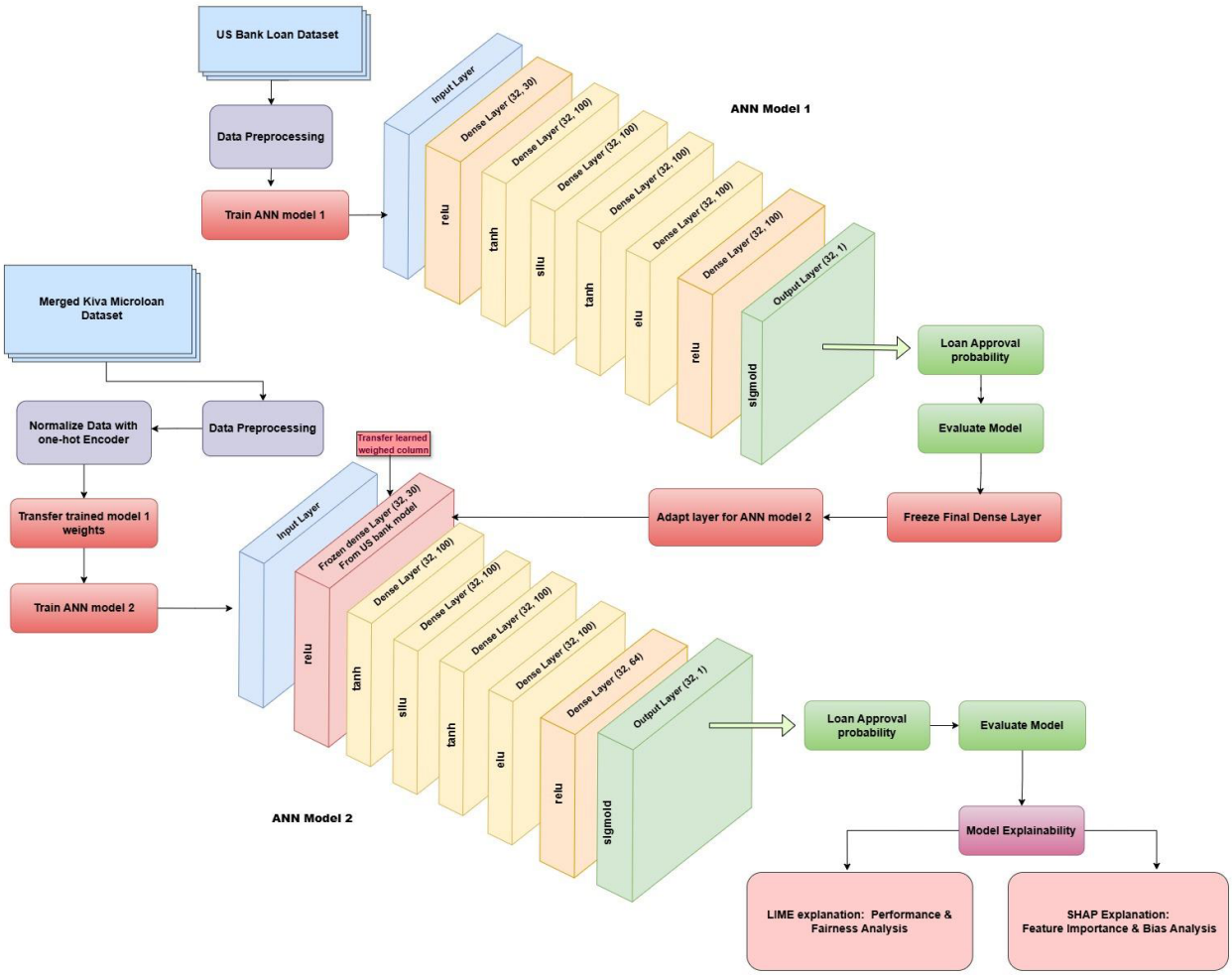

Figure 3.4: Proposed two-stage ANN framework with transfer learning and XAI

Phase 1: The Integration Layer (Merge): The Integration Layer (Merge): Microfinance data were broken down, hence, we collected different data sets depending on Kiva loan records, region MPI indicators and macroeconomic profiles in coun-

tries. The combination of the different sources to one dataset is conducted by the complex pre-processing and hierarchical imputation logic described in Section 3.3. The result of this process is that the thin file applicants are converted into data rich people, with the high environmental and socio-economic context surrounding their applications.

Phase 2: The Unified Risk and Eligibility Architecture: The Unified Risk and Eligibility Architecture: Once data has been gathered, the model is moved to the Unified Assessment phase. Unlike the traditional black-box models that integrate different rejection reasons, this structure divides the decision process into two engines:

- The Eligibility Engine (Deterministic): This layer is a rigid rule (Region is serviceable, and Sector is compliant) which is determined by policies of the institution. It eliminates the applications which are non-viable at a very rapid rate thus operational efficiency.
- The Risk Scoring Engine (Probabilistic): The system measures the likelihood of repayment based on soft signals on the aggregated data. This segregation allows the Microfinance Institutions (MFIs) to modify their risk appetite without altering their fundamental eligibility policies.

Phase 3: The Transfer Learning and XAI Interpretation: The Transfer learning and XAI interpretation: It is a two-stage Artificial Neural Network (ANN) enhanced with Transfer Learning and Explainable AI.

- Source Domain Training: In the first stage, a deep ANN is trained on the data-rich US Bank Loan dataset. Using deep layers with different activation functions and a sigmoid output layer, this model learns general patterns of borrower risk and loan approval trends.
- Target Domain Adaptation: In the second stage, the learned weights are transferred to a new ANN setup for the Kiva Microloan dataset. To handle varying feature distributions, the microloan data is normalized and one-hot encoded. The initial dense layers of the network are frozen to keep the basic knowledge

of risk from the bank data. Additional trainable layers, including a special Adaptability Layer, are added to capture the unique features of micro- finance, like group liability dynamics.

- Explainable Validation: The last ANN provides checks that are comprehensive in terms of loan approval. The SHAP is utilized to validate the universal significance of features and make the arguments trustworthy. LIME is used to offer local explanations to individual decisions. This not only makes the end system accurate and flexible but also understandable and equitable, to help them transfer knowledge among the data-rich banking and data-scarce microfinance situations.

3.6 Summary

The chapter outlines a joint approach based on ANN and XAI to assess the loan eligibility and risk in the developed (US Bank) and developing (Kiva) settings. To this end, a solid Merge pipeline with hierarchical imputation, Feature Engineering, and Transfer Learning and Domain Adaptation protocol have managed to close the data gap in microfinance. Having SHAP and LIME in place guarantees that the risk scores that will be produced will not only be accurate but also transparent and just, which will lead to financial inclusion and responsible risk management.

Chapter 4

IMPLEMENTATION & RESULT ANALYSIS

4.1 Introduction

This chapter offers a critical analysis of the models of artificial neural networks (ANN) developed. It focuses on the transfer learning methodology that predicts the eligibility of microloan based on a strong yet developed country base model. The chapter subsequently explains how this model which has been pre-trained was fine-tuned and trained using the Kiva microloan data. A close comparison of this model with one that has been trained from scratch is done at the end of the chapter. The analysis will contain an analysis using Explainable AI (XAI) frameworks: SHAP and LIME to understand the mechanism used by the model to arrive at decisions. Our structured workflow was based on a guided experimental process, including data set integration (Merge), risk modelling (Risk Eligibility) and interpretability analysis (XAI) in order to achieve consistent and reliable results. All the experiments have been conducted in a Python- based interface (Google Colab) with the use of TensorFlow/Keras as a deep learning platform, Scikit-learn as an evaluation platform, and Matplotlib/Seaborn as a visualization platform.

4.2 Data Integration & Visualization (The Merge Process)

This analysis mainly based on putting together a single, complete microloan dataset. In this section, we describe how different data sources- such as loan records, country details, theme categories, and MPI data- were brought together step by step. The merging process was carried out carefully to fix mismatches and deal with missing values so the final dataset would be clean and usable.

I. Source Datasets & Initial Preparation

The microloan risk and eligibility study relied on a variety of Kiva datasets to gather loan-level, regional and socioeconomic information to conduct their study. These comprised country, loan records, profiles, loan theme labels and regional MPI data. The datasets were loaded individually.

We verified its structure and inspected quality of data. Then the initial cleaning process to eliminate columns which were not required. This reduced repetition. It also simplified processing. The names of columns and data formats were made to fit into a uniform pattern of all datasets. This was done to prevent errors of merging in the future. The combined data had an exact match of the country names, regions and the loan IDs.

II. Multi-Stage Merging Approach

The incorporation of the Kiva datasets was a successful use of systematic multi-stage merging technique:

- **Macro-Economic Integration:** First, the loan dataset was merged with the country profile dataset through the use of the country attribute to add national level economic indicators on each loan record.
- **Thematic Integration:** Loan themes were linked with individual loan IDs to associate certain thematic information with individual loans.
- **Regional Poverty Integration:** Addition of Regional MPI data was done by using a combination of national and regional characteristics. More importantly, the MPI attribute was chosen only in this merge to prevent the duplication of the schema in the different files because the regional naming practises were different.

This level-based method of training prepares a gradual validation at every level reducing the chances of repetitive records and erroneous joins. Accuracy in merging was followed by creating intermediate datasets after each step.

III. MPI Completion & Post-Merge Processing

In order to solve the missing values after the merge, in which case a particular MPI completion procedure was performed. The sources of MPI by regions and location were integrated to form a supplementary MPI reference dataset. The hierarchical logic used to fill in missing values of MPI was: region-level matching and partial text matching in location names followed, and lastly, country-level MPI approximation where needed. Character encoding errors and multi- lingual text were corrected by using normalization techniques. The duplicate entries and residual null values were then removed in post-merge processing which left a clean enriched dataset that was ready to be analysed.

IV. Exploratory Data Visualization

Data visualization after the integration was done to explore feature distributions and trends.

i. Sector Distribution

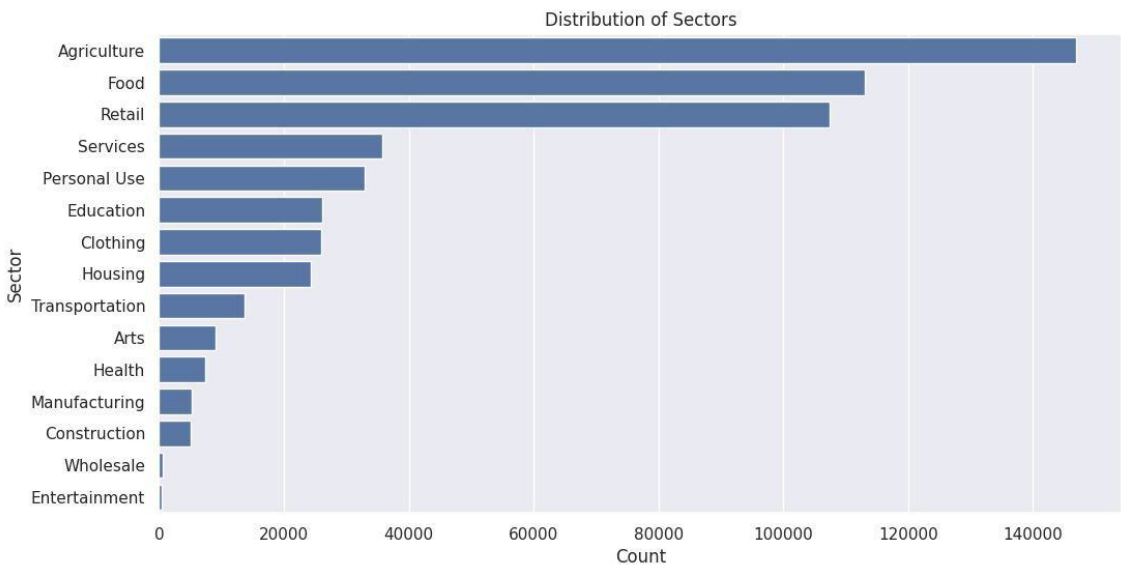

Figure 4.5: Distribution of sectors

The figures in the statistics depicted the fact that the agriculture sector is the most dominating sector with the figure standing at 150,000 loans. Food and Retail are

only other categories that had over 100,000. Under these three top ones, a steep fall comes; intermediate industries such as the Services, Personal Use, and Education industries are expected to lie between 25,000 and 40,000. On the other hand, such industries as Wholesale and Entertainment is underrepresented and it seems that it is the least typical activity.

ii. Top 50 Activities In the count plot, the two most common activities highlighted are Fishing and General Store, the count of which is about 60 000 each. Thereafter after these leaders, there is a significant drop; the third-largest activity, Pigs, has an approximate of 25,000. The rest of the list consists of the combination of the basic retail (Clothing, Food) and the agricultural needs (Livestock, Crops) gradually decreasing to the low-frequency needs such as taxi-services and embroidery.

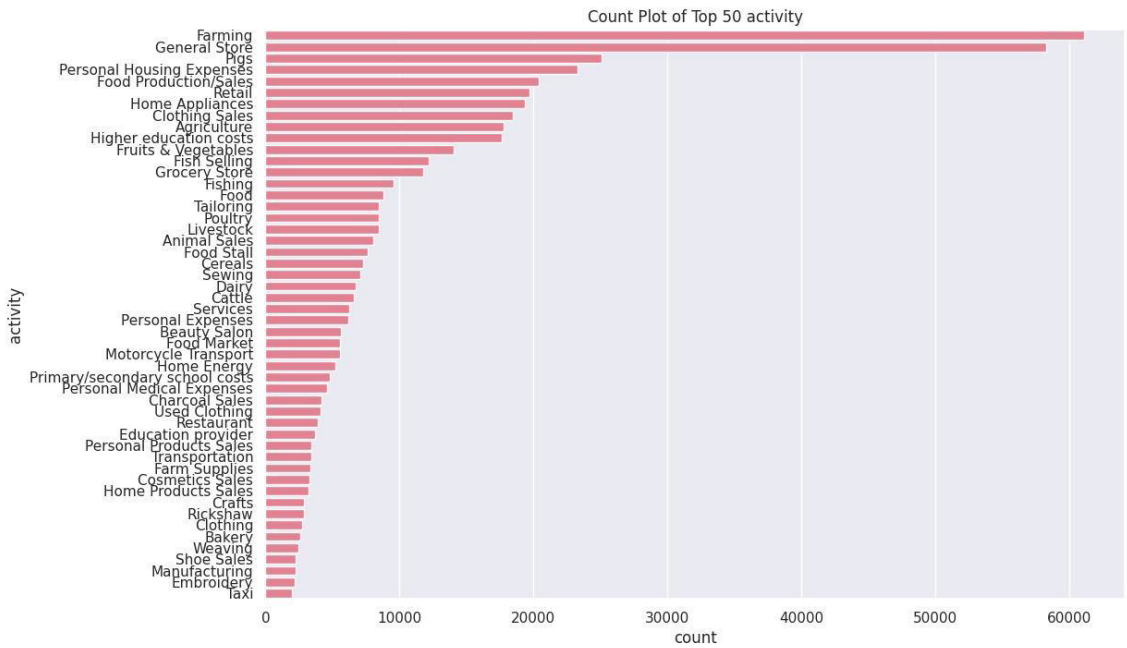

Figure 4.6: Count Plot of the Top 50 Activity

iii. Regional Representation

The most common area in the data set is south-eastern Asia, having over 200,000 entries, almost 2 times more than the second largest area. In second position is Eastern Africa which has nearly 105,000 and thirdly is South America with nearly

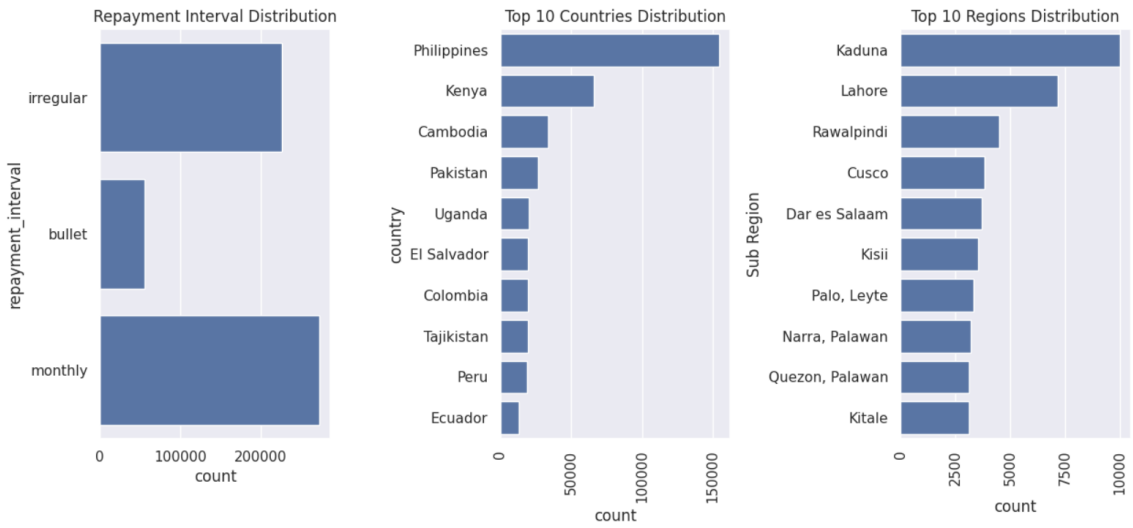

Figure 4.7: Count Plot of Region

75,000. The numbers drop precipitously below these three top ones; the North American continent, and Eastern Europe have very low numbers, Central America and Southern Asia have respectable figures (between 30,000 and 45,000).

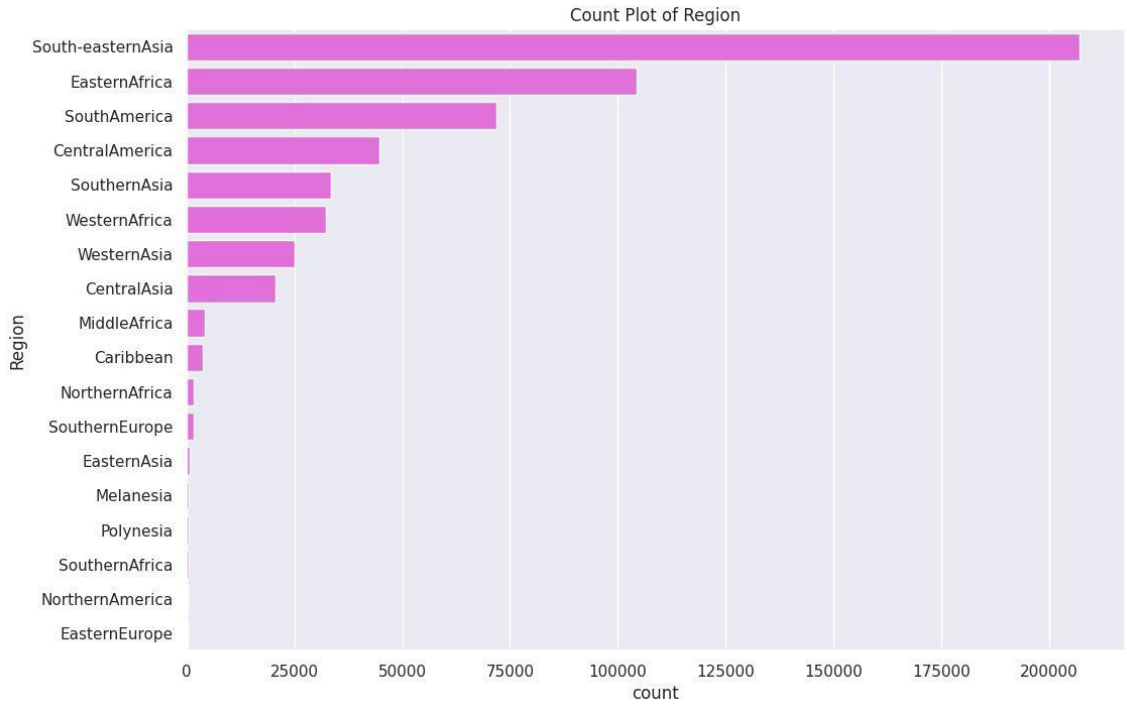

Figure 4.8: Count Plot of Region

iv. Correlation Matrix

The heatmap of the correlation shows the presence of three important relationships:

- **Loan Amount vs. Lender Count:** There is a significant positive correlation (0.82) of this variable meaning that a higher loan amount tends to have more funders.

- **Sectoral Trade-off:** There exists a negative correlation (0.97) between Agriculture and Services employment, which gives a trade-off in the distribution of the workforce.

- **Risk vs. Amount:** The correlation between Risk Score and Loan Amount is positive (0.47). Put simply, bigger loans are perceived to have more credit risk and therefore larger loan values are usually accompanied with more risk-based ratings.

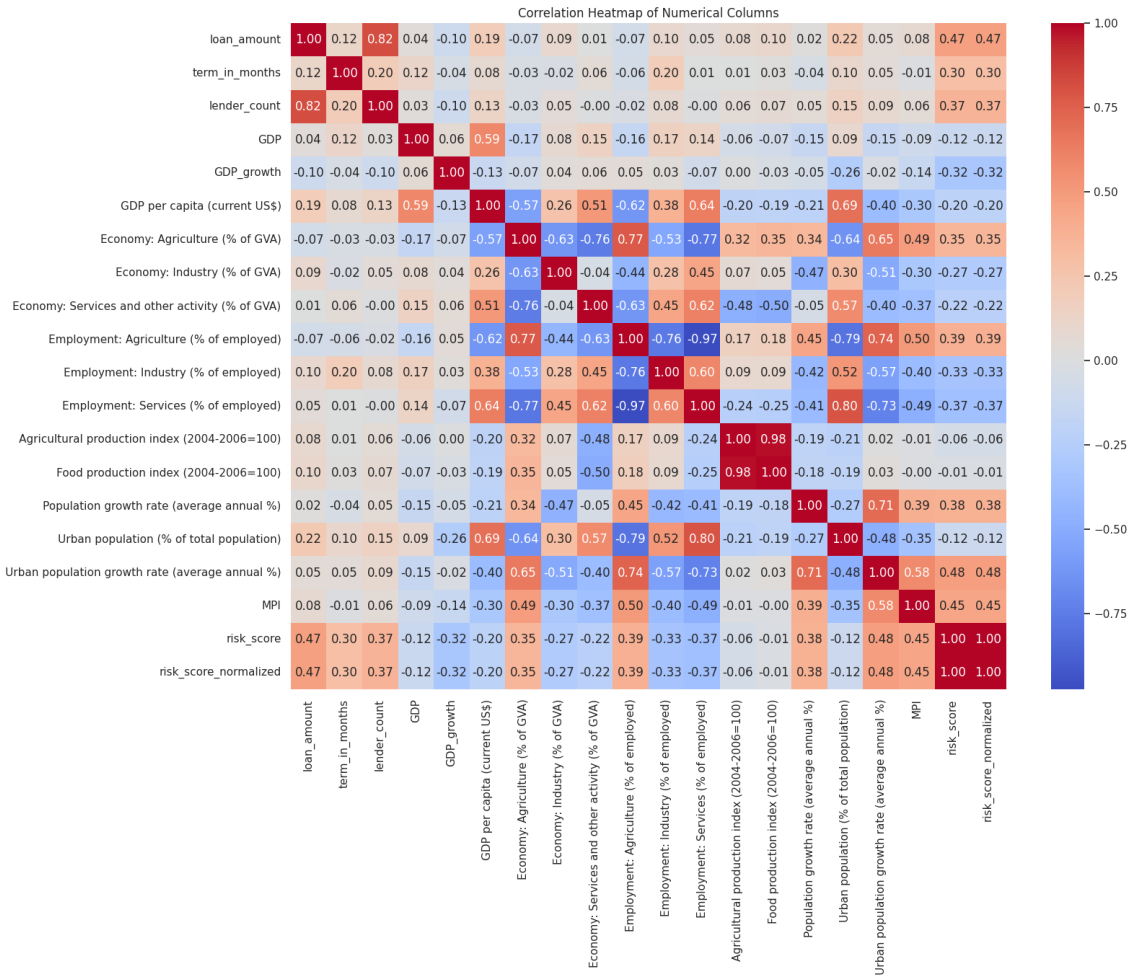

Figure 4.9: Correlation Matrix

v. **Distribution Analysis** The financial data e.g. *loan_amount*, *lender_count*

show a high right skewness, meaning that there are many small loans with many extreme outliers. On the contrary, demographics are relatively steady and follow a normal distribution whereas economic variables, including agricultural labour, become widely and diversely distributed.

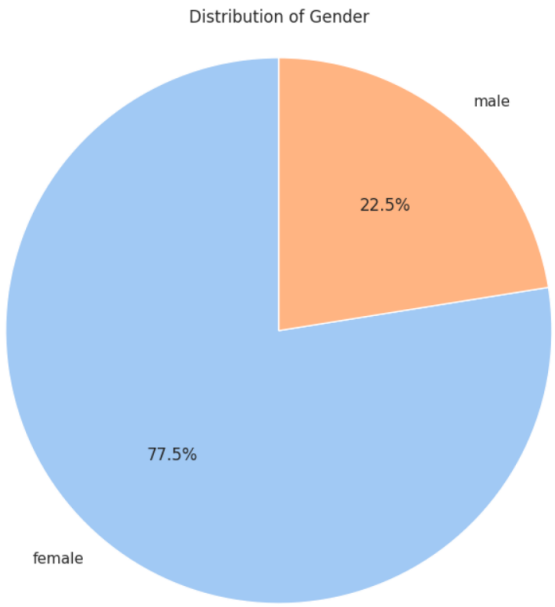

Figure 4.10: Distribution of Borrower Gender

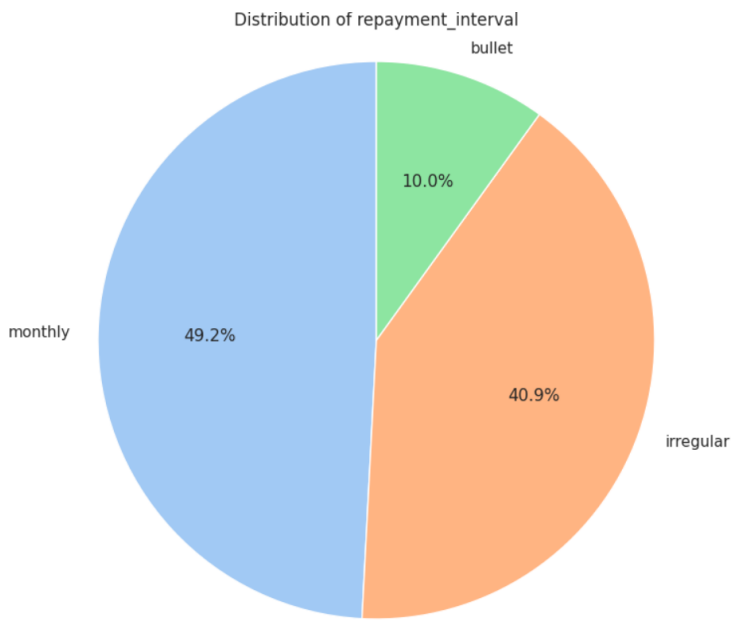

Figure 4.11: Distribution of Repayment Interval

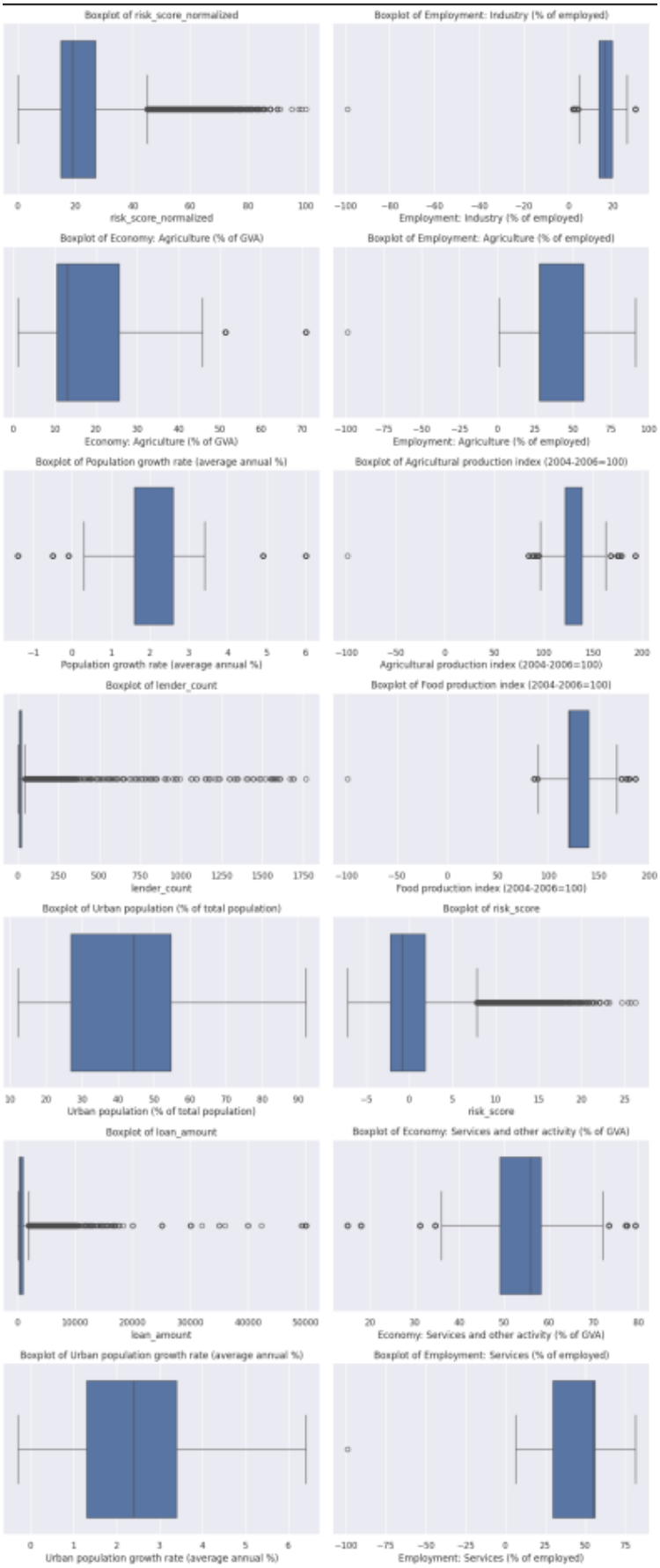

Figure 4.12: Box Plot of different sectors

4.3 Risk & Eligibility Modeling

In this part, the authors present the operational bulb of the study, during which the preparation of the Risk Score, the use of adaptive thresholds, and the functioning of the transfer-learned ANN model are described.

I. Risk Score Construction & Normalization

Risk Score is a composite score, which has been developed to forecast the risk of borrower default through a weighted multi-factor approach. It aims at combining financial, macroeconomic and borrower-level data into one and interpretable value when it comes to loan eligibility.

It has chosen thirteen predictive factors which were classified into borrower demographics, macroeconomic indicators and loan-related characteristics. All of the features were scaled to a one and were given a weight in heuristic fashion depending on their effect on the credit risk:

- **Risk-Enhancing Factors (Positive Weight):** Multidimensional Poverty Index (MPI) ($\omega = +1.0$), Loan Amount ($\omega = +1.0$), and Unemployment Rate ($\omega = +1.5$).
- **Risk-Mitigating Factors (Negative Weight):** GDP per capita ($\omega = -1.0$), Lender Count ($\omega = -0.6$), and Funded Amount ($\omega = -0.4$).

These features were then linearly weighted together to obtain the raw risk score which was then scaled using Min-Max to the range of. Greater scores indicate increased chances of default. The statistical analysis indicated that there is a right skewed distribution with the mean score of 21.73 in the risk category which shows that most of the applicants are moderately creditworthy.

II. Adaptive Risk Threshold Analysis

The quantile-based risk thresholding technique suggested showed strong adapt- ability in a wide range of lending conditions:

- **High-Volume Microloans (USD 100–2,000):** In order to promote financial inclusion, the eligibility criterion was established at the 75th percentile ($\tau = 26.81$). This arrangement categorized 415,542 eligible and 138,514 ineligible borrowers respectively.
- **High-Value SME Loans (USD 1,000–50,000):** A more conservative strategy lowered the mark to the 30th percentile ($\tau = 15.32$). This resulted in 166,184 applicants being eligible and 387,872 rejected.

This is flexible enough to enable the model to range between inclusion-oriented lending approval rates of about 75% and risk-averse capital funding of about 30%.

Feature	**Credit Score**	**Risk Score**
funded_amount	✓	✓
loan_amount	✓	✓
term_in_months	✓	✓
lender_count	✓	×
repayment_interval	×	✓
borrower_genders	✓	×
activity	✓	×
use	✓	×
Loan Theme Type	✓	×
partner_id	✓	×
country	×	✓
region	×	✓
GDP per capita (current US$)	✓	×
GDP growth rate (annual %, const. 2005 prices)	×	✓
Economy: Agriculture (% of GVA)	×	✓
Employment: Services (% of employed)	✓	×
Employment: Industry (% of employed)	✓	×
Urban population (% of total population)	✓	×
Unemployment (% of labour force)	×	✓
Population growth rate (average annual %)	×	✓
Balance of payments, current account (million US$)	×	✓

Table 3: Feature Usage for Credit Score and Risk Score Calculation

III. Training Model Performance (Source Domain)

An initial Artificial Neural Network (ANN) was first trained on the US Bank Loan data (20,000 records, 37 features) before becoming accustomed to the Kiva environment. The feature extractor, which was based on this model of Teacher, was

configured to study the universal patterns of credit risks (like Debt-to-Income ratios and behaviour in payment history) The results of this baseline model were remarkable. The performance of this baseline model was exceptional:

- **Accuracy:** 99.92%
- **AUC:** 0.99
- **Precision/Recall:** Near-perfect scores across all classes.

These almost perfect measures suggest that US banking data have very well organized and deterministic signal to noise ratios. The model did capture the hard financial rationality of a mature economy giving a strong set of pre-trained weights (latent knowledge) that can be transferred to the micro finance setting with limited data.

IV. Micro-Loan Model Performance (Target Domain)

The baseline US Bank ANN was adapted to the Kiva domain by transfer learning. It had five new Dense layers (100, 100, 100, 100, 64) and a Sigmoid output. The model obtained:

- **Overall Accuracy:** 86.95%
- **Validation Accuracy:** 89.52%
- **AUC:** 0.90

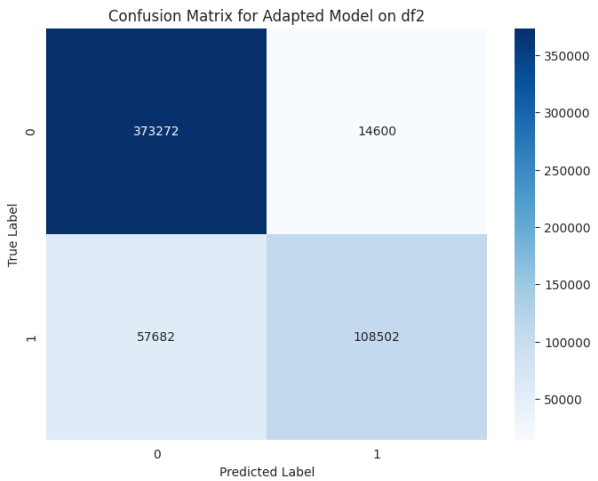

Figure 4.13: Confusion Matrix for Adapted Model on df2

Class	Precision	Recall	F1-Score	Support (n)
Class 0 (Ineligible)	0.87	0.96	0.91	387,872
Class 1 (Eligible)	0.88	0.65	0.75	166,184
Accuracy	0.87			554,056

Table 4: Classification Report (Transfer Learning Model)

While the recall for the Approved class (0.65) indicates a conservative, risk-averse approach, the high precision (0.87) across both classes confirms reliable classification.

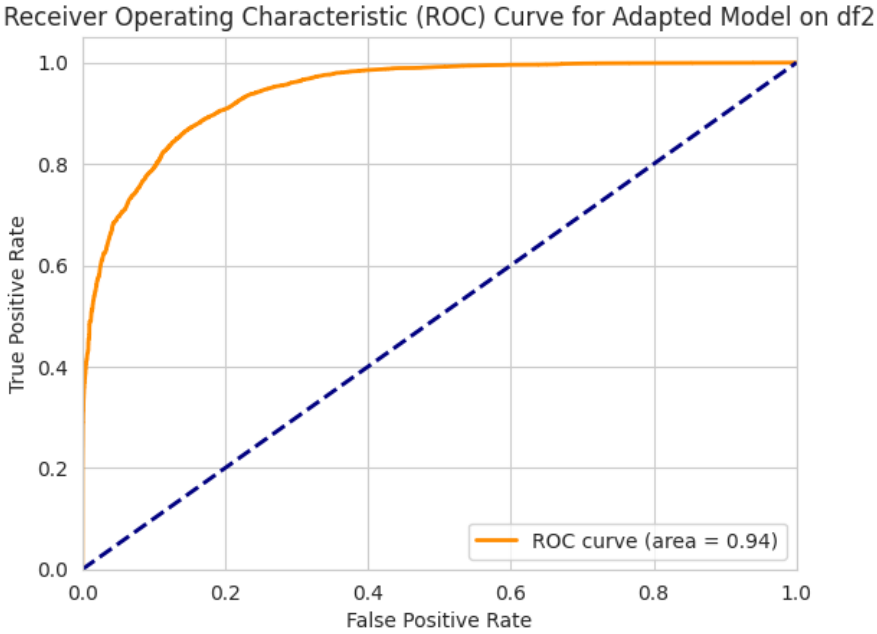

Figure 4.14: Receiver Operation Matrix (ROC) Curve for Adapted Model on df2

V. Transfer Learning Validation

The effectiveness of the transfer learning was confirmed through the comparison of the adapted model with an ANN that was trained on Kiva data only. The

Metric	Baseline Model (Scratch)	Transfer Learning Model (Proposed)	Improvement (Δ)
Validation Accuracy	85.24%	89.52%	+4.28%
Overall Accuracy	82.32%	86.95%	+4.63%
AUC (Area Under Curve)	0.84	0.90	+0.06
F1-Score	0.81	0.86	+0.05

Table 5: Performance Comparison on the Validation Set

transfer learning model achieved higher results on 4.28% higher than the scratch model (Peak Validation Accuracy: 89.52% vs. 85.24%). This substantiates that the

universal financial trends that are acquired through the US dataset were efficiently used to enhance the microloan guesses. microloan predictions.

4.4 Explainable AI and Verification

The model was audited with both global (SHAP) and local (LIME) interpretability metrics in order to guarantee the inclusion of transparency and the analysis of whether the transfer learning process was based on good economic rationale.

I. Analysis of Explainable AI (XAI)

Various techniques of XAI were combined in an interpretation of the black box decisions of the model. Such analyses show that the model is not fuelled by noise of huge financial trends, and confidence is given to the microloan eligibility projections.

II. Local Interpretation (LIME)

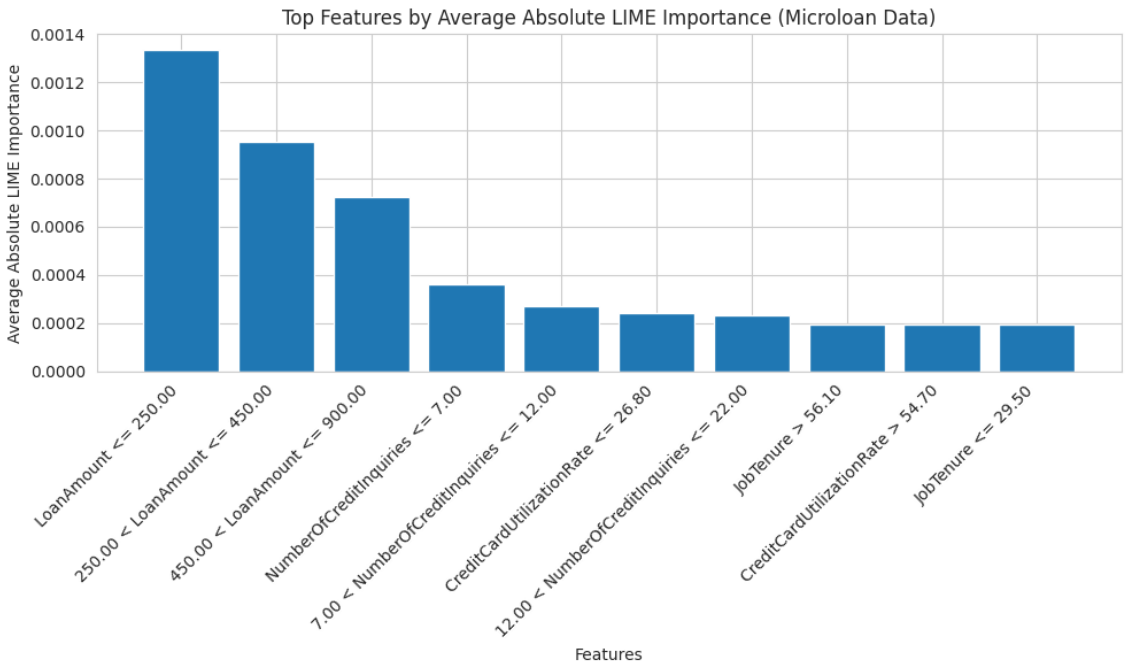

Figure 4.15: Top Features by Average Absolute LIME Importance (Microloan Data)

LIME (Local Interpretable Model-agnostic Explanations) was used to better understand how the model made decisions for individual loan applications:

- **Non-Approved Instances:**Factors like Annual Income and Credit Score had strong negative impacts on the prediction, meaning lower values in these areas pushed the model toward rejecting the loan.
- **Approved Instances:** In contrast, the same financial factors contributed positively when their values were strong, increasing the chances that the loan would be approved.

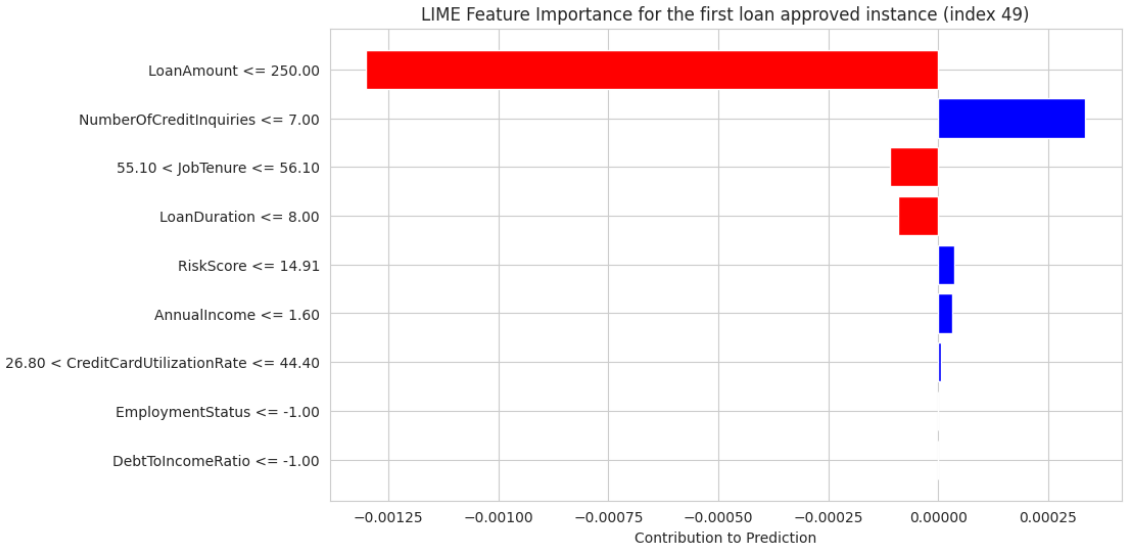

Figure 4.16: LIME feature importance for the first loan approved instance (index 49)

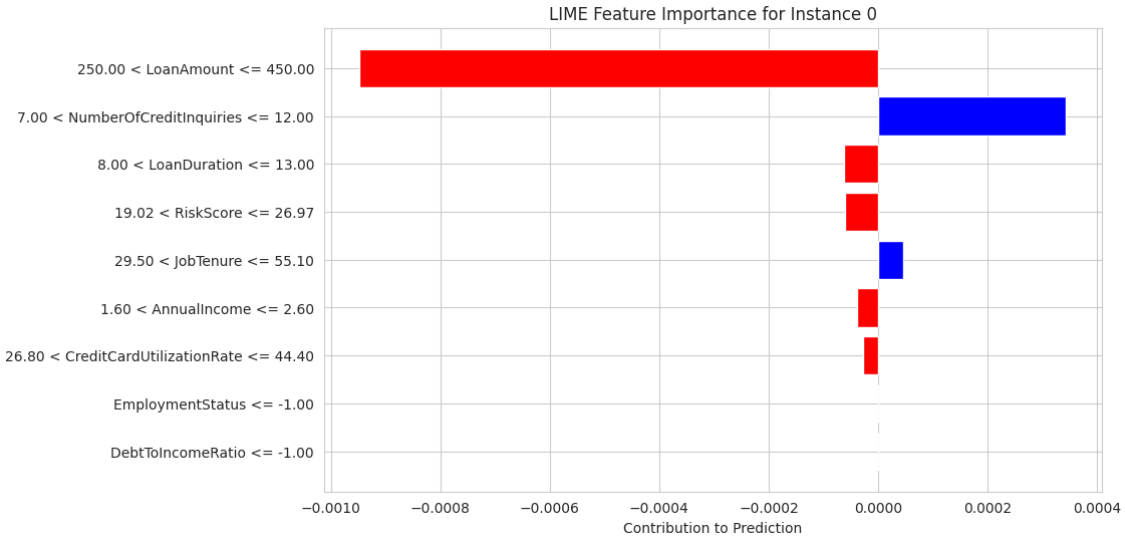

Figure 4.17: LIME feature importance for instance 0

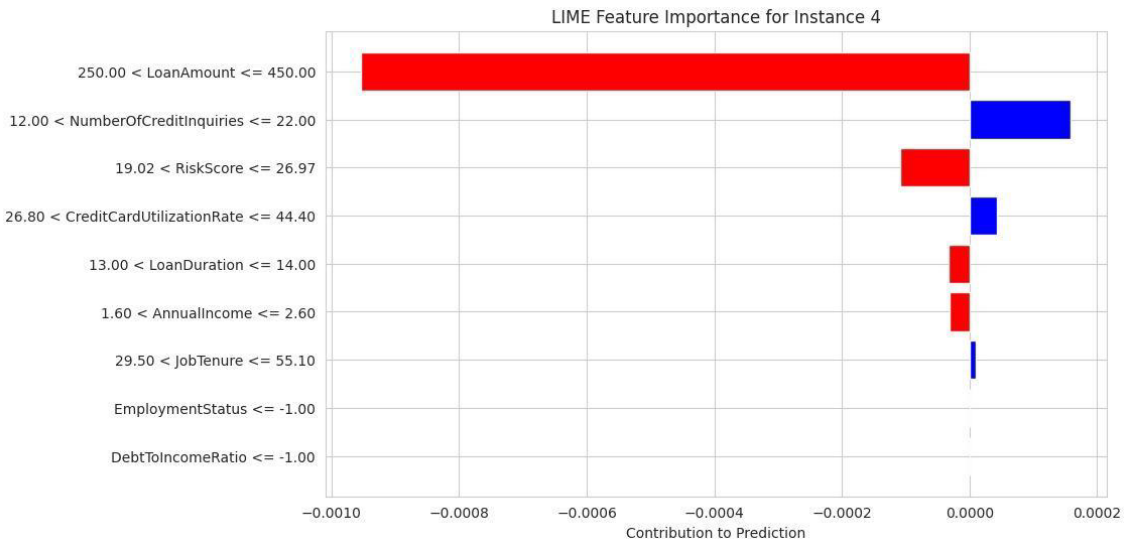

Figure 4.18: LIME feature importance for instance 4

The magnitude and directional behavior of these indicators verify that the model possesses instance-specific reasoning capabilities, confirming that loan approval results are data-driven, comprehensible, and logically consistent.

III. Global Feature Importance (SHAP)

A SHAP Kernel Explainer study of the Kiva dataset reveals that the predictions of the model are largely influenced by few significant insights in the form of financial risk indicators. Risk Score was identified as the most important parameter in the process of loan eligibility and higher scores were strongly associated with non-approval. The second most important was loan amount, which implies that loan request is associated with increased expected risk. The other factors contributed constantly, but insignificantly, such as Job Tenure and Debt-to-Income Ratio. These results facilitate transparency and interpretability in the process of making microloan decisions by assuring that the model is based on relevant and risk-sensitive variables.

A SHAP Explainer study indicated the components of the model being driven globally:

- **Risk Score:** The most prevalent factor; the scores were significantly associated with non-approval.

- **Loan Amount:** The second most influential factor, which suggests that there is an inherent increase in the risk expected when the size of a loan request is large.
- **Secondary Factors:** Aspects like Job Tenure and Debt-To-Income Ratio were found to contribute steadily albeit lesser.

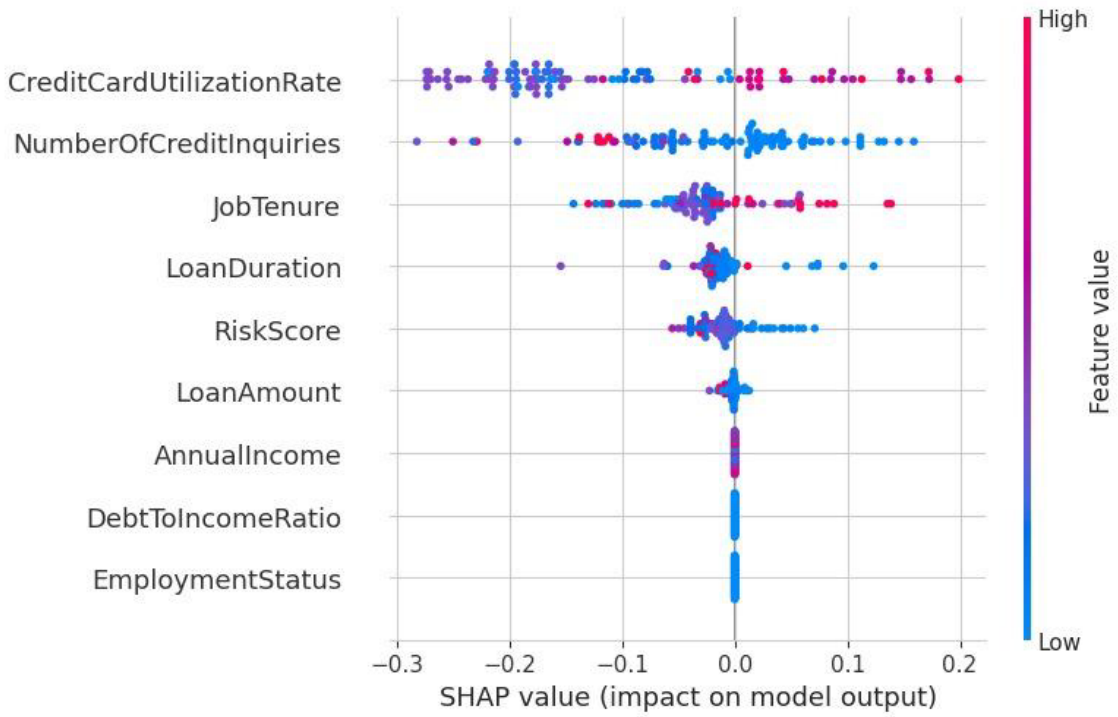

Figure 4.19: SHAP value impact on model output

These findings suggest that it is not only possible to transfer well-developed financial intelligence to the developing regions but also necessary to make credit scoring systems more reliable, flexible, and fair.

4.5 Conclusion

To estimate microloan eligibility, this study built and tested an ANN model that makes use of transfer learning. The model achieved strong discriminative capability, reaching an AUC of 0.90 on the Kiva dataset, along with a validation accuracy of 89.52% and an overall accuracy of 86.95%. These results came from reusing knowledge learned earlier from a US bank loan dataset. Transfer learning offered a 4.28% improvement over a model that show the value of reusing financial risk

patterns. The model's predictions also showed a cautious, risk-aware behavior aimed at reducing possible losses. The model's predictions also showed a cautious, 58 risk-aware behavior aimed at reducing possible losses. It successfully identified 96% of high-risk loan applications, but at the same time it overlooked about 35% of loans that were actually safe to approve. Still, the precision, recall, and F1-score results suggest that the model handled both classes reasonably well and kept the predictions fairly balanced overall. To be interpretable and transparent, explainable AI methods - namely SHAP and LIME were used. These techniques validated that the model largely was based on relevant financial aspects, including Risk Score, Loan Amount, Debt-to-Income Ratio, and Job Tenure, decisions making. The framework incorporates transfer learning, feature engineering, and XAI validation to provide a realistic and comprehensible solution to microloan risk assessment. This would render it an attractive choice in the real lending conditions where clarity, data-driven and responsible decision making are valuable.

Chapter 5

STANDARDS, CONSTRAINTS & MILESTONES

5.1 Sustainability Standards

Data Collection and Use: Open-access and ethically approved finance related datasets, like the traditional banking loan data and the Kiva microloan datasets, are the source of the datasets. To generate a single dataset suitable for multi-layer learning and transfer learning, all datasets are merged, cleaned, and pre-processed. Data accuracy, consistency, and dataprotections are maintained via appropriate processing.

Modelling for Energy Efficiency: By using pre-trained knowledge from traditional loans to microloan data, layered learning and transfer learning reduce the necessity for retraining and processing cost. Repeated processing steps are reduced through efficient feature selection and dataset integration.

Sustainable Implementation: Because the system's processes are low complexity and scalable, it may be applied in organizations with limited infrastructure. Long-term sustainability and minimal resource uses are ensured by effective pre-processing and modular design.

5.2 Impacts On Society

- **Financial Inclusion:** Accurate prediction of loan and microloan eligibility improves credit availability for marginalized groups, enabling small business owners and individuals in developing areas.
- **Transparency and Fairness:** Layered learning in combined with explainable ensure that decisions are understandable to both financial institutions and borrowers, supporting trust and transparency.
- **Economic Growth:** Fare credit availability supports regional economic activity and micro-businesses, encouraging long term community growth.

- **Data Security and Privacy:** To protect the data privacy of the borrower and user confidence, confidential data are anonymized and handled securely processed.

5.3 Ethics

Fairness, responsibility, and data confidentiality are the main ethical principles. All datasets are either publicly available or ethically sourced. Before analysis Private information is deidentified. The decision-making process is made. transparent through the use of explainable AI methods and actions are implemented to limit unfairness and ensure fair predictions within a range of social and economic groups. The method promotes ethical AI use in loan processing, and it is designed to support financial banks instead of taking the place of human judgment.

5.4 Challenges

Data Limitations: Unpredictable data entries in publicly available datasets can restrict the flexibility and consistency of models. This limitation arises due to irregular data entries in publicly available datasets.

Computational Cost: High processing power is required for multi-layered learning and transfer learning along with data integration, especially when handling large datasets.

Time Management: Proper planning is necessary for time-consuming activities, including data cleaning, merging, feature engineering, and performance vali- dation.

Expertise Requirements: Successful implementation and interpretation of model results require interdisciplinary knowledge in data science, AI, and the financial domain. Results require interdisciplinary knowledge in data science, AI, and the financial domain

Regulatory Obstacles: The proposed model refers certain methodologies, and access to data may be limited in order to follow lending and financial data protection regulations.

5.5 Constraints

Data Quality and Balance: To maintain model reliability, data pre-processing and resampling techniques are applied to handle data imbalance and missing data. Infrastructure Constraints: Experiments involving large datasets or complex models may be restricted by a limitations of system resources.

Interpretability vs Accuracy: Even though model performance may be improved by advanced models. Transparency, however, which is important for fair financial decisionmaking, would be reduced.

Privacy Regulations: Data privacy rules must be strictly applied to before sharing or using sensitive borrower information.

Domain Adaptation Challenges: To avoid unfairness and ensure reliable performance, models from traditional bank loans need to be adjusted properly to microloan datasets.

5.6 Gantt Chart and Timeline

The Gantt chart shows the project planning between August 2024 to August 2025 by presenting a step-by-step process from initial planning and data gathering in late 2024 to intensive model development, evaluation, and XAI analysis in early to

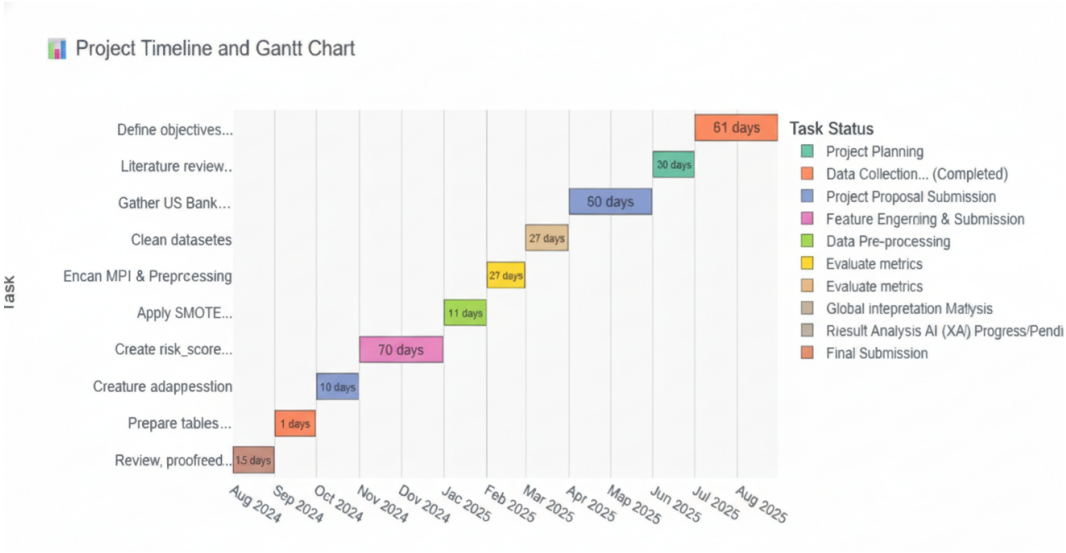

Figure 5.20: Project Timeline and Gantt Chart

mid-2025. A 70-day block in late 2024, a 50-day period in early 2025, and a final

61-day period allocated to the final submission or final reporting stage ending with the project's completion in August 2025 are among the major, lengthy activities highlighted in the calendar.

Chapter 6

CONCLUSION

6.1 Introduction

This paper successfully used a well-organized three step computational framework to analyse loan risk and loan eligibility and has effectively bridged the assessment divide between information abundant banking systems and low resource microfinance settings. The research collected a considerable number of datasets and equalised the regional variations by incorporating a big processing pipeline. One of its innovations was the risk and eligibility structure that implemented exploratory predetermined eligibility limitations and a probabilistic Artificial Neural Network trained through Transfer Learning. The method was found to have excellent predictive power on the microloan dataset of Kiva, outperforming base- line models with a validation accuracy of 89.52% and an overall test accuracy of 86.95%.The systemic application of Explainable AI methods, namely, SHAP and LIME, offers the benefits of transparency in models, which explicitly identifies key financial factors that influence the decision to grant a loan. These findings point to the fact that moving a good financial intelligence to developing markets is not only possible, but it is necessary to make credit scoring mechanisms more reliable, adaptable and just.

6.2 Limitations

Despite the fact that the framework has good performance in some limitation can be considered. Firstly, the Kiva data that is utilized to train the microloan model may not be entirely reflective of the diversity in borrower profile, the financial behaviour or the economic conditions in different areas that are felt in real International lending systems. As a result, the generalization capabilities of the model on new groups of borrowers. The manually assigned risk-score is subjective in nature and may not work across the board. There were also technical problems such as array dimension

inconsistencies when SHAP analysis, the fact that a lot of pre-processing of the data is necessary to align MPI, and tuning of hyperparameters to reduce over-fitting. The issues may affect stakeholder trust and built-in readability. The interpretability remains a factor of consistency of features and data quality even when XAI techniques are used.

6.3 Future Works

This practice can be enhanced in a number of ways in future employment. The capability of the model to generalize between more general financial systems could be enhanced by incorporating datasets that are more geographically and time - heterogeneous. The effect of human intervention can be minimized and model consistency can be improvised, by automatically creating or discovering risk score weightings by using sophisticated optimization or attention- based methods. Making the analysis workflow in a user-friendly application platform which is capable of delivering to the financial institutions transparent and up-to-date loan credit assessment is a very crucial future step in terms of deployment. Developed financial patterns may be found through exploration of developed model architectures, such as ensemble models, hybrid deep learning models, or sequential borrower histories. Lastly, the study of micro-financial organizations to make specific field-validations may help establish the framework as a decision-making tool applicable to microfinance environments by providing a very important information on the issue of operational stability, stakeholders confidence, and real impact.

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